



*Submitted in partial fulfillment of the requirement For the degree of  
Bachelor of Chemical Engineering*

**SYNTHESIS AND CHARACTERIZATION OF CHEMICALLY MODIFIED  
POLYACRYLONITRILE (PAN) FIBER FOR ARSENIC REMOVAL FROM  
SYNTHETIC WATER**

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**NAWABSHAH**

**2026**

## DEDICATION

This dissertation is dedicated to my parents, Guardian & respected Teachers. With the prayer of my parents at every step of my life today I am able for completion of higher degree of Bachelor of Engineering in Chemical Engineering Department. In addition, I am dedicating to my honorable teacher especially to **Engr. Mariam Mir Memon** who always supports and guides me during precious time of my life.

## ACKNOWLEDGEMENT

All praise to Almighty Allah, Whose blessings and Guidance enabled me to complete this research Work titled “ Synthesis And Characterization Of Chemically Modified Polyacrylonitrile (PAN) Fiber For Arsenic Removal From Synthetic Water” this journey has both intellectually enriching and personally Transformative. I am deeply grateful to my supervisor, **Engr. Mariam Mir Memon** For her invaluable mentorship, technical insight, and constant encouragement throughout the course of this project her expertise and dedication were a guiding light at the stage of the research. My heartfelt thanks goes to my group members, Whose cooperation, Hard work, and shared commitment made this endeavor both productive and enjoyable. Their contributions in experimental work, data interpretation and mutual support were essential to our success. I also acknowledge the faculty and staff of the Department of chemical engineering for providing the academic resources and laboratory facilities that made this research possible. Special appreciation to my family and friends for their unwavering support, prayers and motivation during challenging times. This thesis reflects our collective effort to explore sustainable, eco-friendly solutions using locally available resources.

Last but not least, Again, I would like to thanks to my parents and my family for their moral support through my entire life.



## CERTIFICATE

This is to certify that Mr./Miss \_\_\_\_\_ S/o/D/o \_\_\_\_\_  
bearing Roll No \_\_\_\_\_ Final Year Student of Bachelor of Engineering (Chemical) has  
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## ABSTRACT

The existence of arsenic in groundwater has been reported across the world due to its highly toxic nature. It causes a serious public health issue. Various countries across the world are affected by drinking arsenic-contaminated water Pakistan is one of them. Arsenic levels have recently been found to be alarmingly high in the groundwater in Sindh, Pakistan. This study focuses on rural areas of Sindh where drinking water is primarily obtained from the ground. Our research was done by creating synthetic water as sample to perform lab experiments. According to WHO standards of 10  $\mu\text{g/L}$  globally and Standard of EPA 50  $\mu\text{g/L}$ , groundwater arsenic concentrations in affected district of Sindh have reached up to 250  $\mu\text{g/L}$ . In this research, adsorbent polyacrylonitrile fiber was modified by diethylenetriamine, loaded with iron. This chemically Modified polyacrylonitrile fiber adsorbent is used in fixed bed column for arsenic removal for synthetic arsenate water. The characterization of un-Modified and Modified polyacrylonitrile fiber was performed by the FTIR and SEM. The removal efficiency of the fixed bed adsorption column was optimized by changing various operating parameters such as flowrates of 5 mL/min, 10 mL/min, and 15 mL/min, bed height of adsorbent of 1 cm, 2 cm, and 3 cm at pH 5 under equilibrium and dynamic conditions. The Synthetic Arsenate water, was an initial arsenic concentration prepared of 50 g/L. The removal efficiency of synthetic water was achieved 98.5% at a 3 cm bed height and 5 mL/min due to maximum contact time between adsorbent and adsorbate. Sodium hydroxide solution (0.1 M) was found to be capable of regenerating the column bed. The adsorbent, Chemical Modification of polyacrylonitrile fiber, was an effective and efficient adsorbent for the removal of arsenic from synthetic Arsenate water solution.

**Keywords:** Adsorption, PAN Fiber, Synthetic water, Arsenic, Column study.

# **CHAPTER 1**

## **INTRODUCTION**

### **1.1 General Description**

Arsenic in the drinking water has become one of the most concerning environmental problems due to its toxicity and wide distribution. Although arsenic is present in the environment under natural conditions, uncontrolled anthropogenic activities and the discharge of untreated wastewater increase its mobilization, contaminating drinking water sources[1].

Arsenic (As) is one of the most crucial pollutants with strong carcinogenicity. and it can be mainly present in the species of arsenate (As III) and arsenate (As V) in water. Both the health effects and environmental fate of arsenic depend not merely on its total concentration, but on its species [2].

Arsenic is a toxic element (metalloid) linked with a broad variety of neurologic, cardiovascular, dermatologic, and carcinogenic effects, including peripheral neuropathy, diabetes, ischemic heart disease, melanosis, keratosis, and impairment of liver function. Chronic arsenic pollution is now recognized as a worldwide problem, with 21 countries experiencing arsenic contamination of the environment. [3]

Arsenic is a metalloid present in a trace amount in nature. It is found primarily in organic forms such as monomethylarsonic acid, dimethylarsinic acid, and trimethyl arsine oxide, as well as inorganic forms such as arsenite  $As^{3+}$  and arsenate  $As^{5+}$ . In general, inorganic arsenic is more hazardous in comparison with organic forms. Moreover, arsenate is more poisonous as it is more soluble in water than arsenite. The sources of arsenic are volcanic explosions, abrasion of soil, agriculture, industry, dyes, electronic industry, and medical fields. Arsenate binds to the thiol moiety on proteins and enzymes during biochemical reactions and deactivates around 200 enzymes. It can cause neurologic problems, lower RBC counts, resulting in anemia, and lower white blood cell counts, resulting in leucopenia. Exposure to high concentrations of arsenic causes vomiting, diarrhea, and pulmonic hypoxemia. The permissible limit of arsenic in water as recommended by the EPA is 0.01 mg/L[4, 3, 1].

As the population expands and the environment becomes further influenced by climate change, access to fresh and clean drinking water is shrinking. Over 785 million people in the world lack access to clean drinking water. Nearly 800 children die from diarrhea every day due to poor water quality, sanitation and hygiene, water scarcity and sanitation facilities worldwide. Pakistan now faces serious water scarcity and contamination. As the situation gradually worsens, the water crisis has now become

a slogan in the country. In Pakistan, accessibility to clean drinking water is a major human health concern, as the country struggles with water quality and quantity issues. According to the recent global environment performance index (EPI) ranking, Pakistan is among the countries that undergo severe water pollution, water scarcity, poor air quality, global warming, and climate change [5].

Arsenic (IO) A known carcinogen and may lead to lungs, bladder, kidney and prostate cancers Skin diseases, vascular diseases, neurological effects and birth defects Arsenic is ranked number one in the Agency for Toxic Substance and Disease Registry (ATSDR) substance priority list [6].

according to WHO arsenic limit in groundwater is above 10ug per liter in the form of arsenic (III) and arsenic (V); arsenic (III) occurred nearly 60-90 percent as compare to arsenic (V) [6,5,3,7].

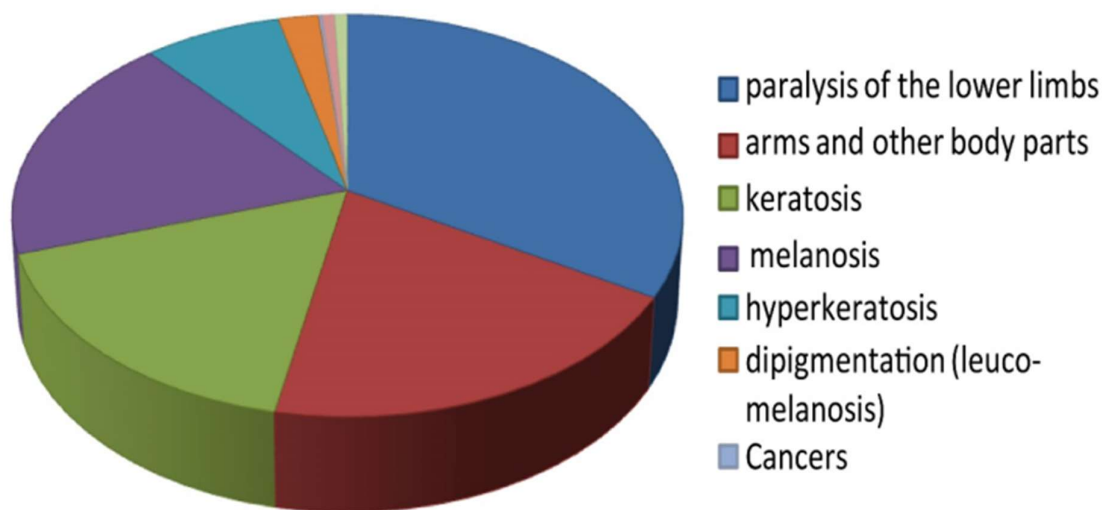
In Pakistan, the appearance of arsenic ground water has an alarming health issue, and Arsenic contamination has emerged as a serious public health concern in Pakistan. In Pakistan's Punjab Province, more than 20% of the population is at risk from arsenic at levels exceeding 10 ppb in groundwater, but just 3% of all people are affected by 50 ppb. The situation is not as good as in Sindh province, where 36% and 16% of people who live there, respectively, consume water that has more than 10ppb and 50ppb of arsenic. Both limited and deep sources are responsible for arsenic pollution [7].

The presence of pollutants (i.e., emerging contaminants, radionuclides, potential toxic metals, and dyes) in water environments has a negative impact on the environment and presents a potential health risk for inhabitants. Among the existing technologies (advanced oxidation process, membrane filtration, biodegradation, coagulation, and flocculation, etc.) for treating pollutants in water, adsorption has garnered great interest due to its low operational cost and fast removal [8].

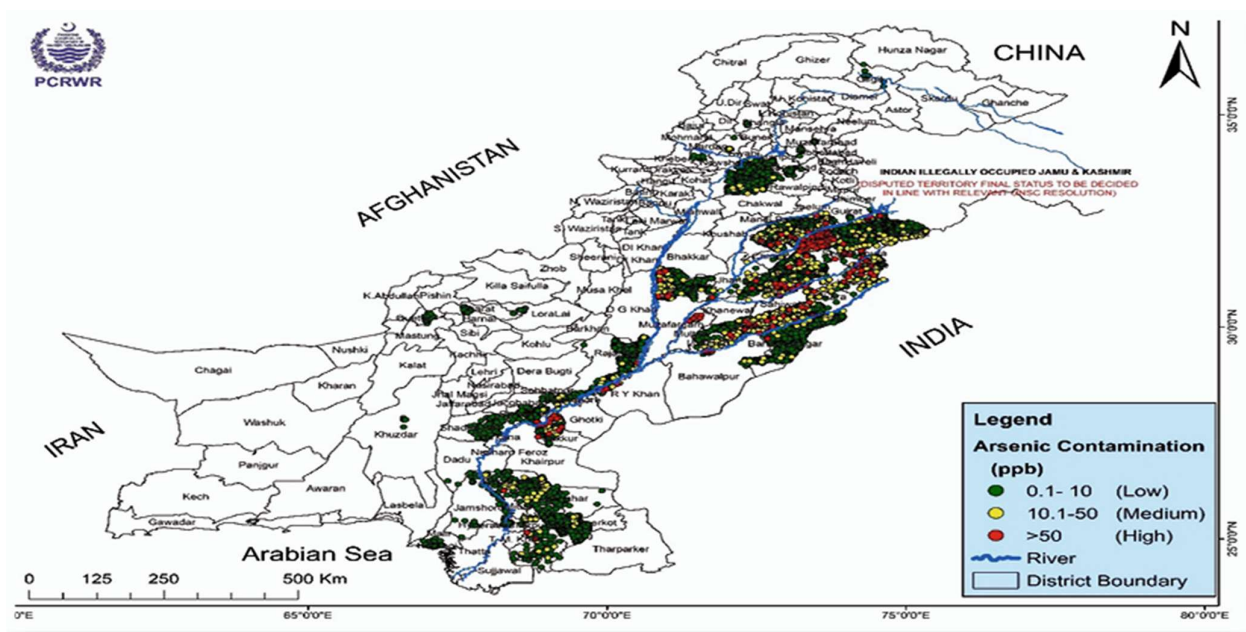
PAN fiber has been the most widely used carbon in the world, so the prepared sample made from PAN fiber shows one of the largest adsorption capacities as a carbonaceous material and adsorption capacities are hardly decreased until equilibrium solution pH (pH) changes from an acidic to neutral range.[7,9,4 ]



**Figure : 1.1 (a) Arsenic poisoning [10]**



**Figure:1.1(b):** Diseases and health issues caused by arsenic [3,4].



**Figure. 1.2 :** Arsenic risk in Pakistan [18]

## 1.2 IMPORTANCE OF WATER

On this planet, water is need for all life; human beings and all other breathing stuffs are incompetent to survive without safe and clean water [7]. Over 2.5 billion individuals worldwide depend on groundwater for consumption, making the provision of high-quality drinking water a critical global challenge [10].

According to the World Health Organization, “safe drinking water” is any water that does not pose a significant threat to health during use, taking into account any variations in sensitivity between different life stages. Since water is the most frequently consumed liquid, it is thought to be the main means of disease transmission. Pakistan is one of the world’s driest regions, and the majority of the people rely on groundwater to meet their water needs for drinking and other household reasons.

Groundwater is a vital resource in Pakistan, providing drinking water and irrigation for agriculture; however, the overuse and overexploitation of this resource have led to a decline in both water quantity and quality in the country. Rising water scarcity and pollution in Pakistan pose a serious danger to the economy and human life, unveiling the importance of the bioeconomy to sustainability.

Water contamination has been a major concern in the country as the population and economy have grown. Water quality assessment is a crucial aspect of ensuring the safety and health of the population. Pakistan faces a number of challenges in terms of providing safe drinking water, including a lack of infrastructure and limited resources. However, there have been several methods used in different studies in the period from 2012 to 2022 to assess the drinking water quality in Pakistan. One of the most commonly used methods is lab analysis [11].



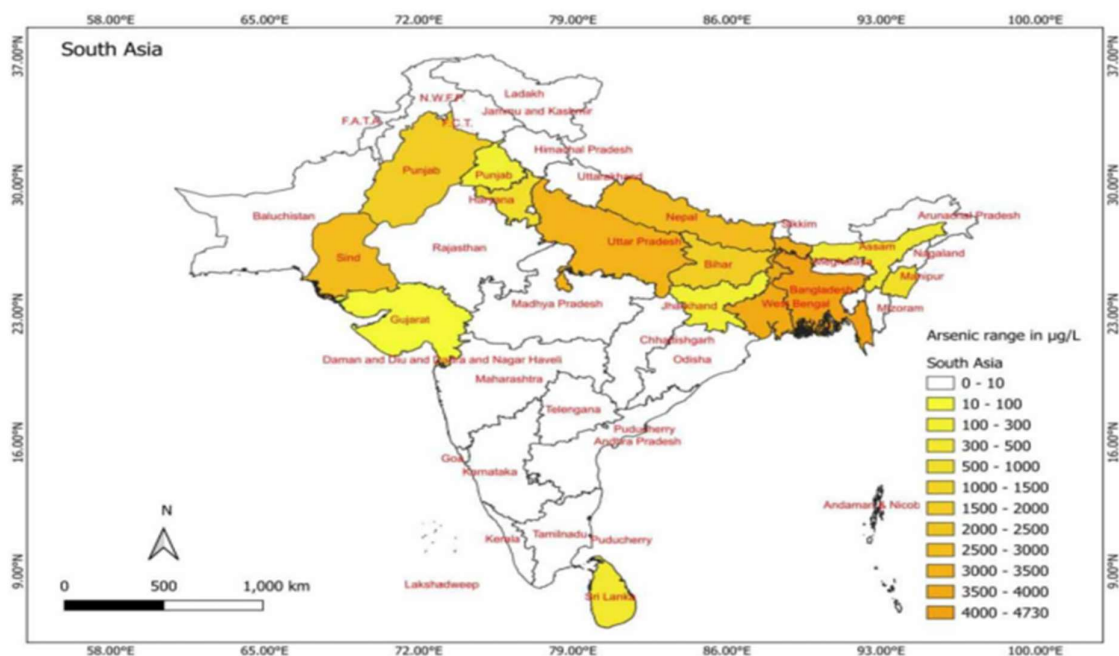
**Fig. 1.3:** Importance of Water and Quality (Science Advanced Journal/WHO Report 2019) [5]

### 1.3. Water bone disease

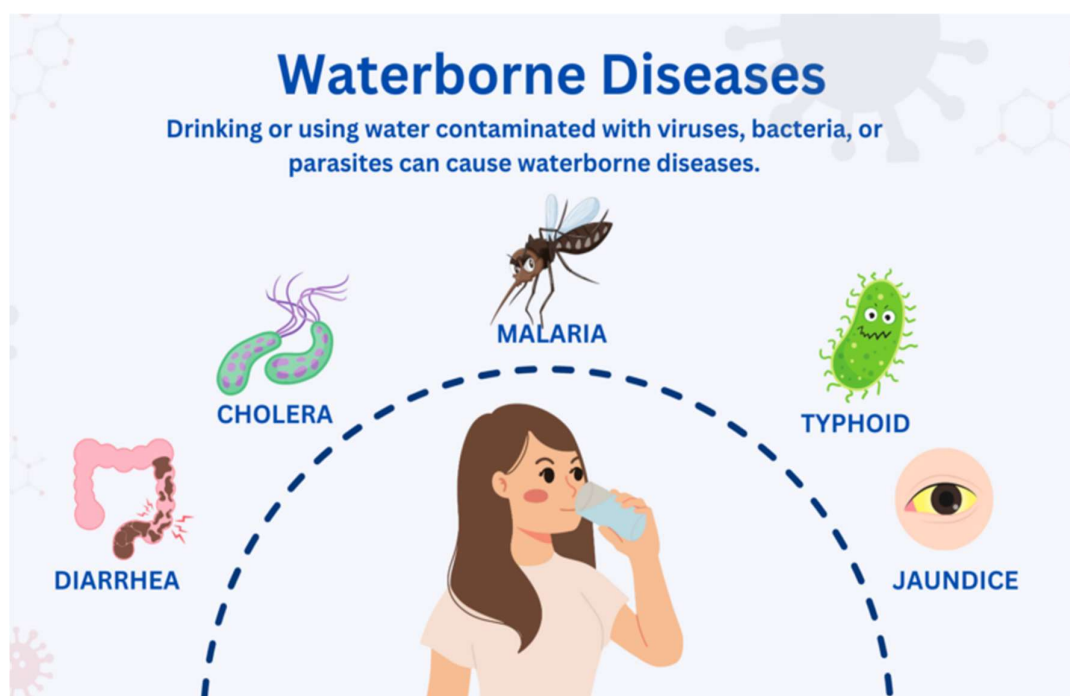
Excessive fluoride intake through drinking water is strongly linked to skeletal fluorosis, a condition that alters bone structure and function. While fluoride at low concentrations supports dental health, chronic exposure above the World Health Organization's recommended limit of 1.5 mg/L can cause pathological changes in bones and joints [12]. The disease is most common in regions where groundwater naturally contains high fluoride levels, including South Asia, East Africa, and parts of China.

Recent literature emphasizes that skeletal fluorosis remains a serious public health challenge. Srivastava and Flora (2020) highlighted that millions of individuals worldwide are exposed to unsafe fluoride levels due to reliance on untreated groundwater [12]. Aryan et al. (2023) reported that in Pakistan and neighboring countries, fluoride contamination often coexists with arsenic, compounding risks of bone disease and systemic toxicity [13]. Clinical outcomes range from mild joint stiffness to severe deformities, with advanced stages characterized by osteosclerosis and restricted mobility [14].

The mechanism of disease progression involves fluoride incorporation into hydroxyapatite crystals, forming fluorapatite. This substitution increases bone density but reduces elasticity, making bones brittle and prone to abnormal remodeling. Children and adolescents are particularly vulnerable, as their developing skeletal systems absorb fluoride more readily. Nutritional deficiencies, especially calcium and vitamin D, further accelerate disease severity [13].



**Figure 1.4** Arsenic distribution (µg/L) in groundwater of various South Asian countries [13]



**Figure 1.5 :** Drinking water contaminants and virus [15]

#### 1.4. Water pollution

In Pakistan, fresh drinking water contamination makes it difficult for people to find clean water supplies. Unhealthy water spreads the prevalence of water borne diseases such as hepatitis, cholera, dysentery, cryptosporidiosis, and typhoid. [5].

Water pollution occurred due to the polluted contaminants which are shown in Fig:1.6 Pakistan map.

A study conducted in arsenic-affected villages in the southern Sindh province of Pakistan highlighted that poor water quality, specifically excessive arsenic levels, was responsible for various skin diseases. By examining the link between arsenic-contaminated drinking water and skin conditions—primarily melanosis and keratosis—the researchers found that individuals who consumed high-arsenic water had significantly higher arsenic concentrations in their hair compared to those who drank low-arsenic water from urban areas. Water pollution is mainly concentrated in industrialization, natural factors, agricultural activities and insufficient water supply and sewage treatment facilities. First, industry is the main cause of water pollution, these industries include distillery industry, tannery industry, pulp and paper industry, textile industry, food industry, iron and steel industry, nuclear industry and so on. Various toxic chemicals, organic and inorganic substances, toxic solvents and volatile organic chemicals may be released in industrial production. If these wastes are released into aquatic ecosystems without adequate treatment, they will cause water pollution. Water is an essential resource for human

survival. According to the 2021 World Water Development Report released by UNESCO, the global use of freshwater has increased six-fold in the past 100 years and has been growing by about 1% per year since the 1980s. With the increase of water consumption, water quality is facing severe challenges. Industrialization, agricultural production, and urban life have resulted in the degradation and pollution of the environment, adversely affecting the water bodies (rivers and oceans) necessary for life, ultimately affecting human health and sustainable social development [15].

## 1.5. Types of water pollution

There are three forms of water pollution, these are briefly discussed below down

- Biological Pollution
- Physical Pollution
- Chemical Pollution

## 1.6. Biological pollution

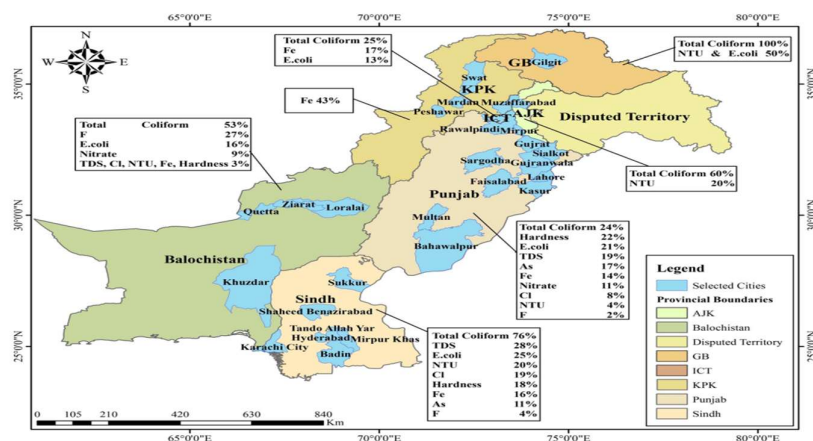
When a variety of microorganisms, such as viruses and bacteria, worms, and/or algae that are present in vast numbers are the main contributors to water contamination. Decomposing organic matter in water, animal waste, and inappropriate human waste disposal all contribute to this sort of pollution.

### 1.6.1. Chemical pollution

Elements or compounds can be contaminants. These pollutants could be created by humans or by natural occurrences. Nitrogen, bleach, salts, pesticides, metals, bacterial toxins, and human or animal medications are a few examples of chemical pollutants.

### 1.6.2. Physical pollution

The primary physical effects of contaminants on water are on its external characteristics. Sediment or organic material suspended in lakes, rivers, and streams due to soil erosion are examples of physical pollutants.



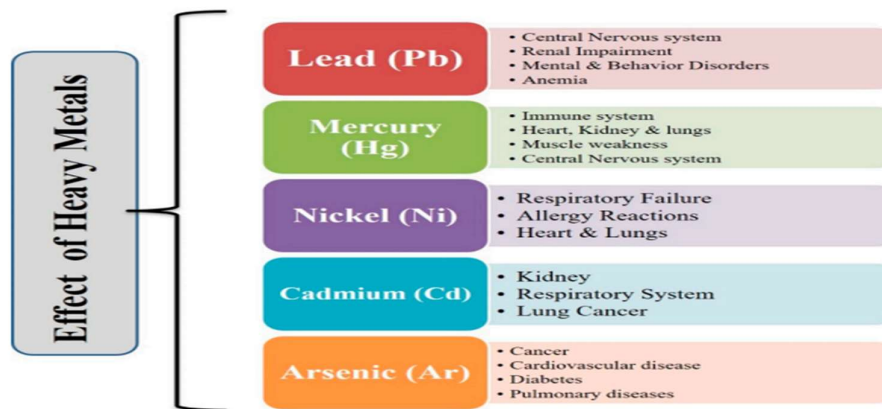
**Figure: 1.6** Pakistan map showing the selected area (blue color) for water quality monitoring and the major contaminants [5].

## 1.7. Heavy metal contamination in water

The presence of toxic substances in aquifers, especially toxic heavy metals, is an important environmental and social concern worldwide. Trace heavy metals, such as iron (Fe), zinc (Zn), copper (Cu), chromium (Cr), nickel (Ni), and manganese (Mn), are major environmental pollutants, particularly in regions with high anthropogenic pressure. These heavy metals are capable of exerting many injurious health effects on human beings by intake of drinking heavy metals-contaminated water. although some heavy metals with low concentrations are essential for life, some metals like As, Cd, Hg, and Pb are biologically non-essential and very hazardous for living organisms[5].

Heavy metals naturally occur in the environment and are vital for survival, but they may become hazardous when they accumulate in organisms. A few of the most frequent heavy metals that contaminate the environment include mercury, cadmium, arsenic, chromium, nickel, copper, and lead [16].

Heavy metals in drinking water are a public health issue. These naturally occurring elements are toxic even at low concentrations and can accumulate in the body over time, a process known as bioaccumulation. Their presence in water supplies stems from both natural sources and human activities like industry and agriculture, raising global health concerns.



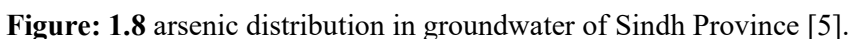
**Figure: 1.7:** Effects of heavy metals on the health of living system [4].

Heavy metals present in water are enlisted below.

1. Arsenic
2. Lead
3. Mercury
4. Cadmium
5. Chromium

In water, its permissible limit is 10 ppb, even though its higher concentration is found in mineral resources. Arsenic is a metalloid present in a trace amount in nature [4] Arsenic (As) is one of the most crucial pollutants with strong carcinogenicity. And it can be mainly present in the species of arsenite (AsIII) and arsenate (AsV) in water [2]. In water, arsenic mainly exists in two oxidation states: As<sup>3+</sup> and As<sup>5+</sup>. Both are highly toxic and pose serious health hazards [17]. Long-term exposure to As<sup>3+</sup> in drinking water can cause life-threatening chronic diseases such as malignancies and cardiovascular disease.<sup>8</sup> As<sup>3+</sup> is 60 times more toxic than As<sup>5+</sup> in drinking water and is one of the most toxic pollutants present in our surroundings and the environment. Arsenic is well known to cause long-term toxicity that has severe health consequences, including malignancies, such as lung cancer, bladder cancer, skin cancer; cardiovascular diseases, and diabetes [18].

More than 150 million people worldwide are involved in arsenic-related health problems. People in Pakistan, Vietnam, Nepal, China, India, Myanmar, and Cambodia drink water contaminated with arsenic [19]. Arsenic, cadmium, and chromium are vital pollutants discharged in wastewater, and the industrial sector is a significant contributor to harmful pollutants [15].



### 1.8.1. Arsenic compounds and their occurrence

It has an atomic number of 33 and is a member of group VA of the modern periodic table, arsenic belongs to the same group with phosphorous and nitrogen. Naturally, arsenic can exist in +5 (arsenate), +3 (arsenite), 0 (arsenic) and -3 (arsine) oxidation state [17].

Much of the arsenic in the atmosphere comes from high-temperature processes such as coal-fired power plants. In relation to anthropogenic sources, mining, metal smelting and burning of fossil fuels are the major industrial processes that contribute to arsenic contamination of air, water, and soil. and burning vegetation, but also from volcanic activity [3].

### 1.8.2. Inorganic arsenic compounds

Inorganic arsenic impairs male reproduction by reducing the weights of the testes, the accessory sex organs, and the number of sperm in the epididymis. Aside from affecting sperm production, inorganic arsenic exposure also causes variations in testosterone and gonadotropin levels, as well as disturbances in the steroidogenesis process [16].

In general, inorganic arsenic is more hazardous in comparison with organic forms. Moreover, arsenate is more poisonous as it is more soluble in water than arsenite. The sources of arsenic are volcanic explosions, abrasion of soil, agriculture, industry, dyes, electronic industry, and medical fields[4].

### 1.8.3. Organic arsenic compounds

It is found primarily in organic forms such as monomethylarsonic acid, dimethylarsinic acid, and trimethyl arsine oxide, as well as inorganic forms such as arsenite  $\text{As}^{3+}$  and arsenate  $\text{As}^{5+}$ . [4] The removal of organic matter from wastewater presents a significant challenge due to its persistent and harmful nature, often requiring expensive and complex treatment processes. Fiber adsorbents have attracted considerable attention for their remarkable efficiency, cost-effectiveness, and potential for reuse [20].

### 1.8.4. Arsenic in the environment

Arsenic exists in the form of various chemical species differing by their physicochemical behavior, in toxicity, bioavailability and biotransformation. The determination of arsenic species is an important issue for environmental, clinical and food chemistry [3].

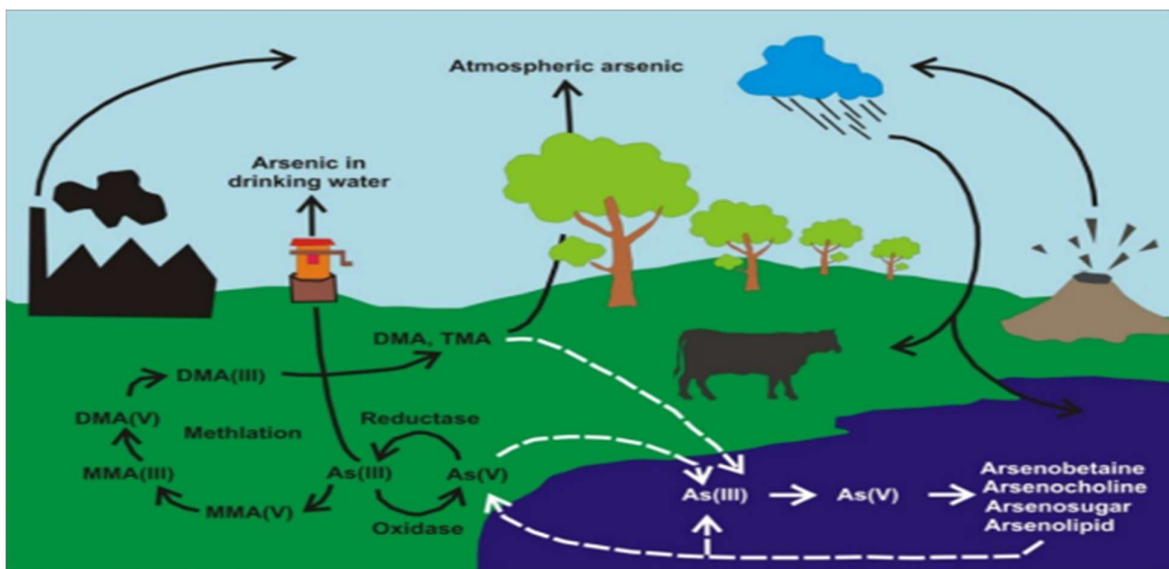
**Arsenic** = Arsenic is stable in the air and is emitted by both natural and anthropogenic sources. The natural sources include volcanic eruption, emission from soils or sediments by microbial reduction, dispersion of As containing particles by wind, evaporation from arsenic compounds and marine organism. anthropogenic sources are high-temperature processes such as smelting of non-ferrous metals, burning of fossil fuels, vegetation and wastes, etc. Arsenic in air is present mainly in particulate

forms as inorganic As and released into the atmosphere primarily as oxides that adsorb on particulate matter (PM), which are dispersed by the wind to remote areas. The particulates then settle down both by dry and wet deposition. Approximately two thirds of the atmospheric As is of anthropogenic origin [3].

**Water.** Water receives As from natural deposits and from both agricultural and industrial effluents. It is found in the atmosphere, soils and rocks, natural waters, and other organisms. As is mobilized in the environment through a combination of natural processes and anthropogenic activities, e.g., atmospheric dry and wet depositions, weathering reactions, biological activity and volcanic emissions, mining activity, combustion of fossil fuels, use of arsenical pesticides, herbicides, livestock feed additive, wood preservatives, etc [3].

**Soil.** The background levels of As depends on soil types, ranging from 5–10 mg kg<sup>-1</sup>. The soil environment is contaminated with As by natural and anthropogenic sources. Exposure to soil in which the presence of unwanted chemicals/substances higher than the normal concentration disturbs the health of living organisms [16]. Arsenic in soils results from anthropogenic activities, and the main sources in the soil are naturally occurring minerals, such as arsenopyrite, gelignite, realgar, orpiment, etc [3].

**Sediment.** As is also present in the sediment bound to clay minerals and Fe, and Mn oxides, As is a redox sensitive element, present in the sediments of surface water reservoirs, formed by continuous weathering and erosion [3].



**Figure: 1.9** The different mechanism controlling the mobility of arsenic in environmental component [3].

### 1.8.5 Environmental Arsenic Impacts On Human Health

Arsenic is distributed extensively in the atmosphere, most often in combination with oxygen (O), chlorine (Cl), and sulfur (S). In the environment, arsenic is largely associated with sulfide minerals. Enargite ( $\text{Cu}_3\text{AsS}_4$ ), cobaltite ( $\text{CoAsS}$ ), loellingite ( $\text{FeAs}_2$ ), realgar ( $\text{AsS}$ ), orpiment ( $\text{As}_2\text{S}_3$ ), mispickel ( $\text{FeAsS}$ ), niccolite ( $\text{NiAs}$ ), and tennantite ( $\text{Cu}_{12}\text{As}_4\text{S}_{13}$ ), are the most prominent arsenic bearing minerals [17]. the effects of arsenic human health are described in ( figure : 1. 10 )

Exposure to environmental toxicants down-regulates the production of antioxidant enzymes such as superoxide dismutase, catalase, and glutathione peroxidase and increases oxidative stress. In response, ROS are generated, harming the lipids, proteins, carbohydrates, and DNA in cells [21].

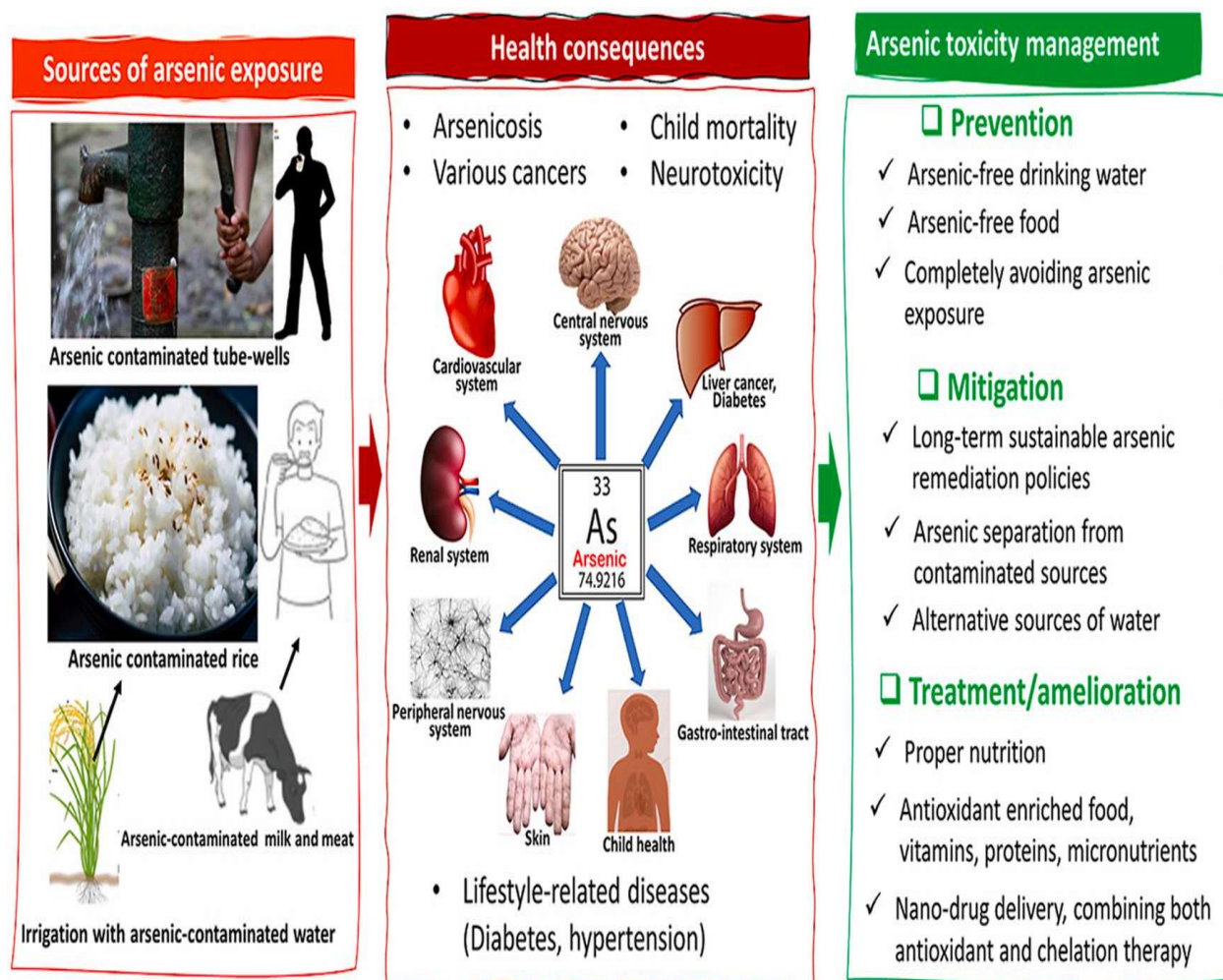


Figure: 1.10: Impacts of arsenic on human system [3].

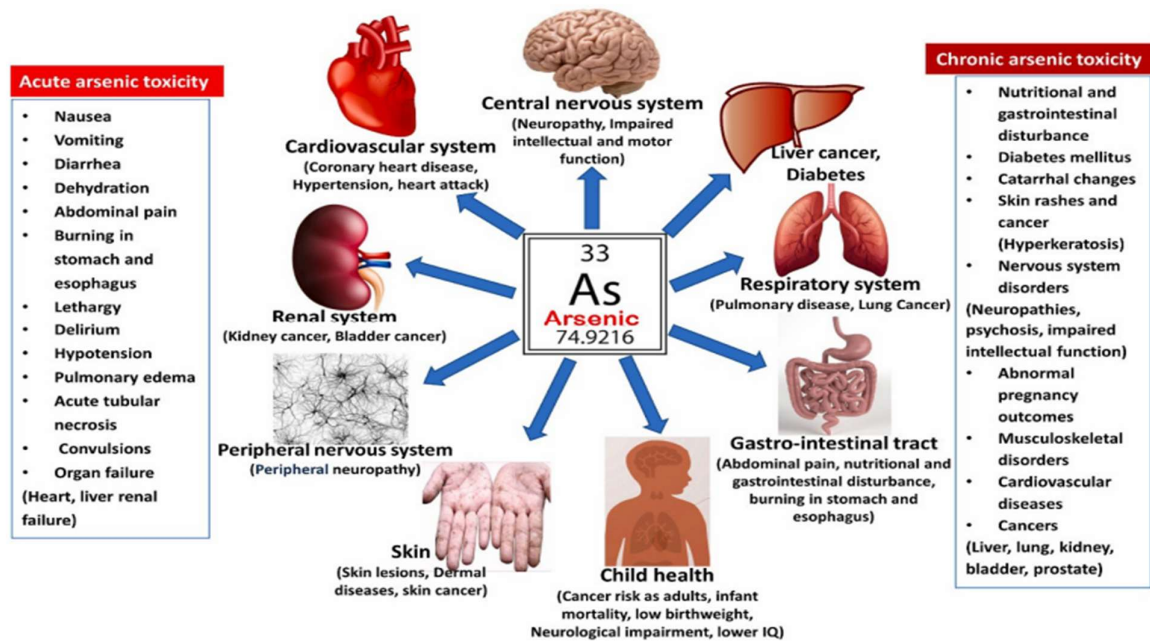


Figure 1. 11 Effects of arsenic on human health [22].

## 1.9. Arsenic removal technologies

The technologies for purifying water that has been contaminated with arsenic are the main subject of this study. The different known arsenic removal techniques are discussed in below and shown in (Figure:1.12) including ion exchange, coagulation/flocculation, phytoremediation, oxidation, adsorption, bioremediation, and membrane techniques including electrodialysis and ultra-filtration. [22]



Figure: 1.12 different arsenic removal techniques [22].

### **1.9.1 Ion-Exchange**

Ion exchange is method in which the same signs is swapped reversibly between solid and liquid in a highly insoluble solution. Ion exchange chromatography is an ion exchange technique that is popular because of its high capacity and resolving power [23]. The ion exchange method is used for water purification (especially converting hard water into soft water) and also for other purifications. It can detect inorganic salts. Cationic exchangers and ionic exchangers are two types of ion exchange. Two techniques are used for ion exchange. One is batch method and second is column method [22]. Ion exchange resins are of two types, namely, cation exchangers and anion exchangers. There are several research papers available on ion exchange resins that are used to extract toxic ions and dyes [4].

### **1.9.2 Coagulation/Flocculation**

Arsenic contamination water is treated using coagulation, which is then followed by the neutralization and filtration processes. Various arsenic species are transformed into flocs for further filtration using various coagulants. [23] The coagulation process is also preferable because of its simplicity and removal efficiency. This process is applicable on small scales and large scales, but pH adjustments are obligatory [22].

### **1.9.3. Phytoremediation**

Phytoremediation is a cost-effective biological technique that can easily safeguard human health and the environment from the poisonous effects caused by arsenic. It is a procedure in which plants accumulate pollutants from the soil [22].

### **1.9.4. Oxidation**

The arsenate As(III) form of arsenic is highly versatile and needs conversion into a less mobile form, As(V), because most of the treatment procedures viably eliminate just As(V); that is why pre-oxidation of As(III) to As(V) is needed and, for this purpose, different oxidation processes like Fenton's reagent, hypochlorite, and permanganate. and oxidizing agents like ozone, chlorine, bleaching powder, and hydrogen peroxide are used [22].

### **1.9.5 Adsorption**

The adsorption process is an inexpensive method having a very low operating cost. Furthermore, it causes less toxicity during the extraction process of hazardous metals compared with other conventional methods such as filtration, sedimentation, coagulation, chemical precipitation, solvent extraction, neutralization, etc [4]. Adsorption is the phenomenon where molecules of an adsorbate substance attach to the surface of an adsorbent substance through intermolecular forces. Adsorption has been widely used as a wastewater treatment technique for removing contaminants from aqueous solutions, such as organic pollutants, heavy metals, dyes, nutrients, and pathogens [24]. Adsorption is

a process in which molecules bind (adsorbate) individually to the surface of a solid material (adsorbent) without penetrating its internal structure. Adsorption can take several forms: physical, chemical, and functionalized surface adsorption [10] Adsorption is a versatile technique widely haracte in wastewater treatment for the efficient removal of a diverse range of pollutants, encompassing both inorganic and organic compounds [23] This method is characterized by its high adsorption capacity and rapid kinetics [20]. Adsorption is an exsitu technique used to remove heavy metals, e.g., arsenic [22]

### **1.9.6 Bioremediation**

Arsenic is hazardous to health—not only to human health, but it is also dangerous for marine life when it is present in water. Now it extends around the globe, so bioremediation is a technique used to remove arsenic. Bioremediation is a helpful technique to remove arsenic from water, soil, and Mine tailings [22].

### **1.9.7 Membrane Technology**

it is a membrane structure that serves as a barrier, forcing contamination to stop as molecules flow through it. A porous membrane serves as the material for the barrier. Nano-filtration, micro-filtration, ultra-filtration, and electrodialysis are the four basic forms of membrane technology[22]. membrane separation is one of the approaches that is most affordable and practicable[25].

### **1.10 Polyacrylonitrile Fiber (PAN)**

Polyacrylonitrile fiber has many favorable characteristics, such as elasticity, huge surface area, and excellent toughness, making it a super adsorbent. The polyacrylonitrile structure contains semi-crystalline and amorphous phases. Polyacrylonitrile contains a large number of carbon – carbon in its formula, which is responsible for its stability and excellent toughness. It is used to produce carbon nanofibers owing to its high carbon content (56%), and thus, stable products are formed through the reaction of the nitrile group [4]. The utilization of synthetic fibers, particularly polyacrylonitrile fiber (PANF), has garnered significant attention due to their convenient and economical properties, wide spread accessibility, and the abundance of nitrile groups that can be readily transformed into various functional groups such as carboxyl, amide, amidoxime [20]. PAN fiber has been the most widely used carbon in the world, so the prepared sample made from PAN fiber shows one of the largest adsorption capacities as a carbonaceous material and adsorption capacities are hardly decreased until equilibrium solution pH (pH) changes from an acidic to neutral range[7].

The aim of the present study is to use modified Polyacrylonitrile (PAN) Fiber as adsorbents for the removal of Arsenic. To evaluate the performance of the adsorption column and modified PAN fiber. [7]) as the Polyacrylonitrile are shown in figure: 1.13



**Figure: 1.13** Un-Modified PAN fiber [19].

### **1.12. Problem statement**

Formerly, the arsenic in groundwater is the main contributor to health issue (Cancer) in Pakistan. A survey of the literature indicates that, most of the ground water is used without any proper treatment that negatively impact on our health. Therefore, Adsorption technique is recommended to produce safe drinking water. However, there are some major problems related with adsorption systems were identified by many researchers.

1. Adsorbents are limited in their application to remove arsenic ions from real contaminated water due to their lower capacity and recycling.
2. Adsorbents for arsenic removal include activated alumina, iron, iron oxide, Aluminum oxide, titanium oxide, and mixed metal oxides, has been used. However, there is still a strong challenge in developing a highly efficient and economical adsorbent for arsenic removal.
3. Therefore, novelty of this work is the modification and functionalization of Polyacrylonitrile Fiber which can increase the arsenic removal efficacy thereby decrease the process cost.

### **1.13. Scope of work**

This study focused on the polyacrylonitrile (PAN) fiber, which was chemically modified by introducing functional groups.

1. The modified fiber was analyzed through Fourier Transform Infrared Spectroscopy (FTIR) and Scanning Electron Microscopy (SEM) for characterization.
2. The modified PAN was used in column experiments for arsenic removal.
3. Atomic Absorption Spectroscopy was conducted to analyze the adsorption capacity of the modified PAN fiber.

### **1.14. Objectives**

1. To develop chemically modified polyacrylonitrile (PAN) fiber
2. To characterize the chemically modified PAN fiber
3. To analyze and evaluate the efficiency of Chemically Modified PAN fiber for Arsenic ( $A^{+5}$ ) removal through column study

## CHAPTER 2

### LITERATURE REVIEW

**(Memon & unar et al., 2023)** investigated arsenic adsorption using a packed bed column designed with chemically modified polyacrylonitrile (PAN) fiber. The adsorbent was prepared by coating PAN fiber with chitosan/iron oxide, and its performance was evaluated through simulation and modeling studies using Aspen Adsorber V11. Both small-scale (2 cm diameter, bed depths 1–3 cm, flow rates 1–3 mL/min) and large-scale (8 cm diameter, bed depth 8 cm, flow rates 512–1536 mL/min) columns were tested [7].

**(Chen & Li et al. 2025)** reviewed recent advancements in adsorption and detection of wastewater pollutants using covalently functionalized polyacrylonitrile fibers (PANF). PANF, widely used in textiles and carbon fiber production, can be chemically modified by transforming nitrile groups into functional groups (amines, carboxyl, amidoxime, quaternary ammonium salts). These modifications enhance adsorption capacity for heavy metals such as  $\text{Hg}^{2+}$ ,  $\text{Fe}^{3+}$ ,  $\text{Cu}^{2+}$ ,  $\text{Ni}^{2+}$ ,  $\text{Pb}^{2+}$ ,  $\text{Cd}^{2+}$  and  $\text{Cr}^{3+}$ , as well as organic pollutants and dyes [20].

**(Tran, H.N., 2023)** compiled a Special Issue on adsorption technology for water and wastewater treatments, highlighting 12 research papers from international contributors. The studies focused on developing advanced adsorbent materials for removing contaminants such as antimony (Sb), chromium (Cr(VI)), copper (Cu(II)), zinc (Zn(II)), fluorine, phenol, dyes (indigo carmine, Congo red, methylene blue, crystal violet), and pharmaceutical pollutants (levofloxacin, captopril, diclofenac, paracetamol). The studies applied batch adsorption experiments under varying operational conditions (pH, contact time, adsorbent dosage, concentration, temperature, stirring speed). Adsorption kinetics, isotherms (Langmuir, Freundlich, Redlich-Peterson, Toth, Sips, etc.), and thermodynamic analyses were widely used. Several works also employed machine learning models (KNN-GWO, Gaussian regression with particle swarm optimization) and statistical physics approaches to predict adsorption capacity and optimize process parameters [8].

**(Kaveh & Jazani et al., 2024)** investigated the design, preparation, and characterization of modified PAN-based carbon fibers using an anodic electrochemical oxidation process. The study focused on the role of electrolytes in altering fiber morphology and surface properties to improve adhesion in composite applications [9].

**(Patel & Pandey et al., 2023)** presented a comprehensive review on arsenic contamination in the environment, focusing on its sources, mobility, exposure pathways, and analytical detection methods.

Arsenic is recognized as a persistent pollutant with severe neurological, cardiovascular, dermatological, and carcinogenic effects, making it a global health concern [3].

**(Alam, E., 2025)** provided a comprehensive overview of metal–organic frameworks (MOFs) for arsenic remediation, highlighting their structural tunability, large surface area, and adjustable pore sizes as key advantages for water treatment. MOFs are crystalline porous materials composed of metal ions/clusters coordinated to organic ligands, enabling selective adsorption of contaminants such as arsenic [18].

**(Cheng & Zhang et al., 2022)** developed phosphonic acid functionalized polyacrylonitrile (PPAN) hollow fiber membranes via a Mannich reaction to enhance heavy metal adsorption. The study focused on  $\text{Hg}^{2+}$  removal from aqueous solutions, evaluating adsorption capacity, kinetics, isotherms, and reusability [25].

**(Kumar & Kumari et al., 2023)** conducted a comprehensive review of water quality assessment and monitoring in Pakistan, highlighting contamination levels, monitoring frameworks, and public health implications. The study synthesized findings from 2012–2022, covering surface water, groundwater, and rainwater sources. Literature review of chemical, physical, and biological monitoring methods used in Pakistan. Data collected from Google Scholar and Scopus for the period 2012–2022. Comparative analysis against WHO and national drinking water standards [11].

**(Dilpazeer & Munir et al., 2023)** presented a comprehensive review of arsenic contamination in groundwater, its sources, health impacts, and the latest removal technologies. Arsenic is recognized as a Class-1 carcinogen by WHO, with global exposure affecting more than 300 million people, particularly in Asia [22].

**(Shokri & Naghashzargar et al., 2024)** developed adsorptive electro spun nanofibrous membranes (AENMs) based on polyacrylonitrile (PAN) incorporated with Cloisite 20A (C20) organoclay for arsenic ( $\text{As(V)}$ ) removal from contaminated water. The study investigated structural properties, adsorption performance, and regeneration ability of the fabricated membranes [26].

**(Satyam & Patra, 2024)** presented a comprehensive review of adsorption-based wastewater remediation, highlighting historical developments, recent innovations, and persistent challenges. The study emphasizes the role of nanotechnology, advanced fabrication methods, and AI/ML integration in designing next-generation adsorbents for sustainable water purification [24].

**(Lan & Reina et al., 2023)** compared three low-cost technologies for arsenic removal from water at high concentrations (5 mg/L): zerovalent iron ( $\text{Fe(0)}$ ), coagulation-flocculation, and adsorption on

natural clay. The study focused on both As(V) and As(III) species, evaluating efficiency, kinetics, and practical applicability [23].

**(Gupta & Sharma et al., 2022)** reviewed the role of polyacrylonitrile (PAN) as a versatile and cost-effective polymer for fabricating adsorbents used in wastewater treatment. The paper highlights PAN's structural properties, functionalization methods, and applications in removing heavy metals and dyes from contaminated water. where in the methodology. Forms of PAN adsorbents: ion exchange resins, chelating resins, fibers, membranes, and hydrogels. Synthesis techniques: suspension polymerization, emulsion polymerization, redox polymerization, grafting, and electrospinning. Functionalization: PAN modified with ligands such as amidoxime, polyethylene imine, acrylamide, thiazoline, amino acids, and thiourea to enhance selectivity and adsorption capacity[4].

**(Zheng, Liwei. 2021)** investigated reusable electro spun carbon nanofiber composites for the selective removal of inorganic arsenic species (As(III) and As(V)) from water. The study focused on developing high-performance, regenerable adsorbents using nanofiber technology. Where methods are used, Material preparation: Electrospinning of carbon nanofibers combined with functional additives to enhance arsenic adsorption. Characterization techniques: SEM (morphology), FTIR (functional groups), XRD (structural analysis), BET (surface area), and contact angle (hydrophilicity). Adsorption studies: Batch and dynamic experiments for both As(III) and As(V) [2].

**(Rahman & Uddin et al., 2025)** reviewed the role of bio-based nanomaterials in groundwater arsenic remediation, focusing on mechanisms, challenges, and future perspectives. The study highlights eco-friendly, cost-effective nanostructures derived from biological sources as sustainable alternatives to conventional adsorbents [10].

**(Fida & Li et al., 2023)** reviewed the status of drinking water contamination in Pakistan and its associated human health risks. The study synthesized data from 2010–2022, including government and NGO reports, to highlight pollutants, sources, and health impacts. Where Systematic review of 21 bacteriological studies, 19 physicochemical/heavy metal studies, and 29 health complication report. Geographic coverage: 29 major cities across Punjab, Sindh, KPK, Balochistan, GB, ICT, and AJK.[5].

**(Shah & Arjunan et al., 2023)** reviewed the major physicochemical and biological contaminants in drinking water and their associated human health risks, with comparisons between WHO guidelines and national standards (UK, USA, Canada, Pakistan, India). The study emphasized the importance of monitoring and regulatory frameworks to ensure safe drinking water. Literature review of 59 research articles and regulatory reports (2019–2020) from the UK Drinking Water Inspectorate (DWI), WHO, EPA, Health Canada, and CAWST. Focused on contaminants repeatedly failing compliance standards

in England and Wales. Comparative analysis of water quality standards across five countries versus WHO guidelines [6].

**(Mendoza-Chávez & Carabin et al., 2021)** investigated the optimization of electrocoagulation (EC) systems for arsenic removal from synthetic water and applied the optimized conditions to real arsenic-contaminated groundwater in Sonora, Mexico. Where Experimental setup: Bench-scale electrocoagulation reactor (1.7 L volume) with monopolar electrode arrangement (iron or aluminum anodes, stainless steel cathodes). Design approach: Response Surface Methodology (RSM) and factorial design to evaluate effects of current intensity, treatment time, and electrode material. Synthetic water: 100 µg/L As(V) prepared from sodium arsenate. Real groundwater: Collected from Potam village, Sonora, Mexico (initial As concentration ~80 µg/L, pH 8.4, conductivity 1746 µS/cm). Pilot-scale design: Proposed reactor to supply 1 L/s for a community of 6417 people [1].

## **CHAPTER 3**

### **MATERIALS AND METHODOLOGY**

#### **3.1. Overview**

This chapter presents the materials and methods employed in conducting the research, explained comprehensively. It is divided into three sections: the first outlines the chemical modification and functionalization of polyacrylonitrile fibers along with their characterization; the second details the adsorption study of the prepared adsorbent for arsenic removal from synthetic arsenate solutions; and the third focuses on the parametric investigation of adsorption and the evaluation of isothermal adsorption behavior.

#### **3.2. Experimental study**

##### **3.2.1 Materials and reagents**

Bulk materials Polyacrylonitrile fiber was purchased from USA, Diethylenetriamine (DETA), Sodium Hydroxide (NaOH), Aluminum Chloride Hexahydrate ( $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ ), Hydrochloric Acid (HCL), ferric chloride hexahydrate ( $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ ), Acetic acid ( $\text{CH}_3\text{COOH}$ ) were purchased commercially from market of standard grade and used throughout the study.

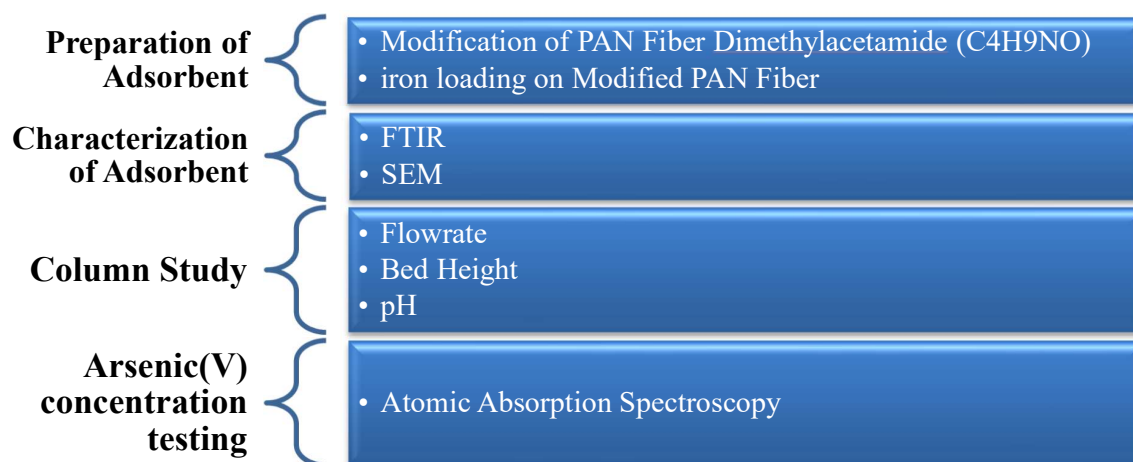
Analytical grade chemicals, reagents and gases were used throughout the study are as shown in Table 3.

**TABLE 3.1** Analytical grade chemicals, reagents and gases are shown in this Table below.

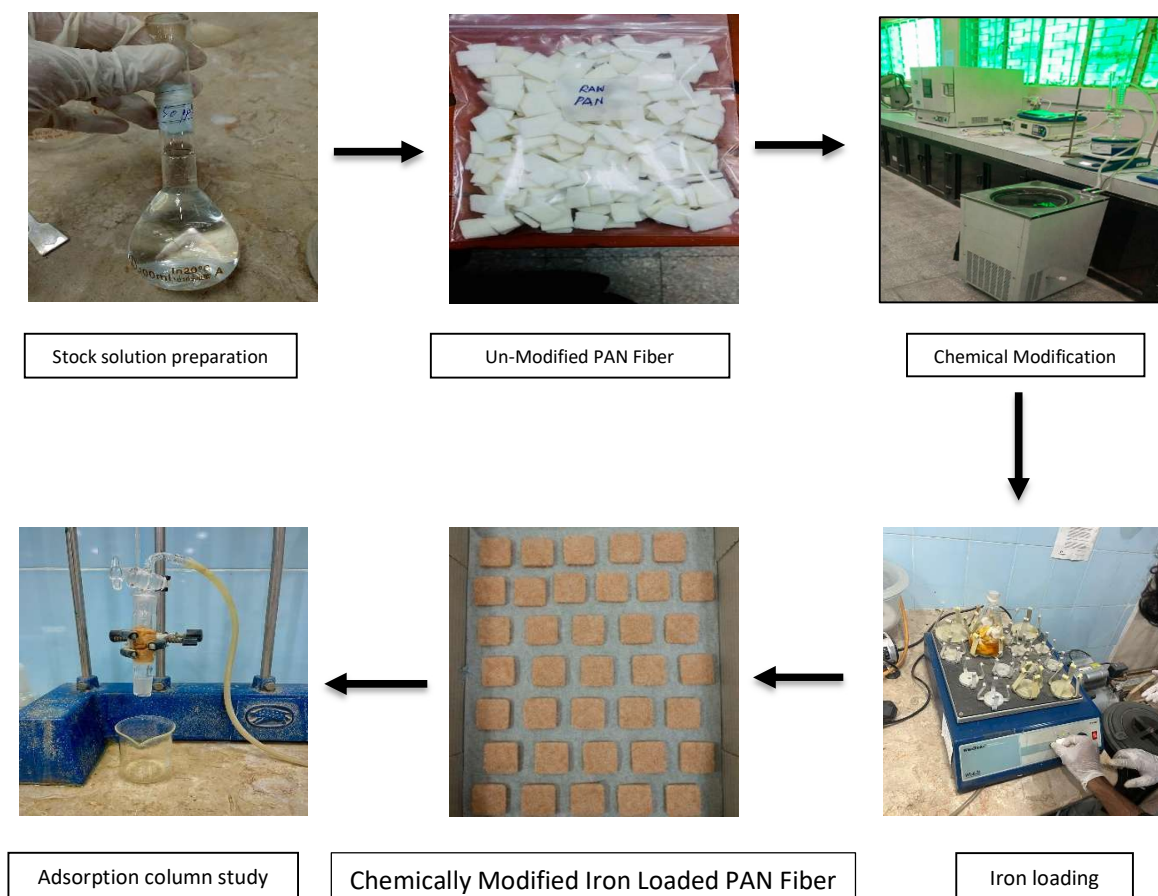
| Reagents                                                                          | Supplier                   | Purity      | Equipment's                                    |
|-----------------------------------------------------------------------------------|----------------------------|-------------|------------------------------------------------|
| Polyacrylonitrile fiber                                                           | Sigma Aldrich, USA         | $\geq 90\%$ | Heating Mantle                                 |
| Potassium arsenate monobasic( $\text{H}_2\text{AsKO}_4$ )                         | Sigma Aldrich, USA         | $\geq 90\%$ | pH Meter                                       |
| Sodium Hydroxide ( $\text{NaOH}$ )                                                | Sigma Aldrich, USA         | 96%         | Agitator/Stirrer                               |
| Hydrochloric acid ( $\text{HCl}$ )                                                | Sigma Aldrich, USA         | 35-37%      | Mechanical Shaker                              |
| Diethylenetriamine ( $\text{C}_4\text{H}_{13}\text{N}_3$ )                        | Daejung chemicals, Korea   | $>97\%$     | Atomic-Absorption Spectroscopy                 |
| Nitric acid ( $\text{HNO}_3$ )                                                    | RCI Labscan Ltd., Thailand | 65%         | Scanning Electron Microscope (SEM)             |
| Iron (III) chloride-6-hydrate( $\text{Cl}_3\text{Fe} \cdot 6\text{H}_2\text{O}$ ) | Daejung chemicals, Korea   | $>98\%$     | Fourier-Transform-Infrared Spectroscopy (FTIR) |
| Sulfuric acid ( $\text{H}_2\text{SO}_4$ )                                         | RCI Labscan Ltd., Thailand | 98%         | Energy-dispersive X-ray analysis (EDX)         |
| Aluminum chloride hexahydrate ( $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ )       | Daejung chemicals, Korea   | $>97\%$     | Electric oven                                  |
| Acetic acid ( $\text{CH}_3\text{COOH}$ )                                          | Sigma Aldrich, USA         | 98%         | Water shaking bath                             |
|                                                                                   |                            |             |                                                |

### 3.2.2 Schematic Diagram Of Methodology

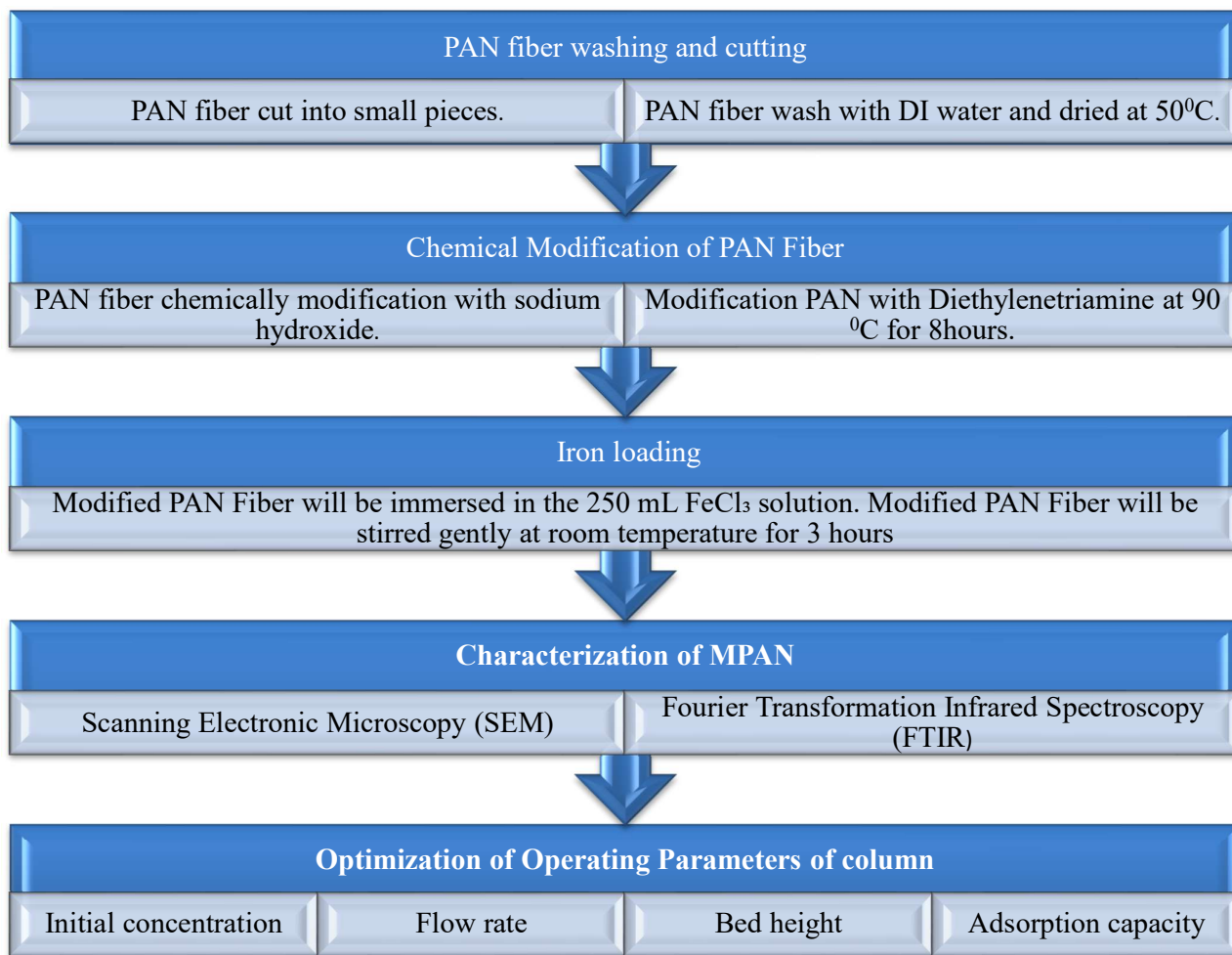
The overall work has been divided into the following phases of research studies.



#### Pictorial View of Methodology



**Figure:3.1** Pictorial view of Methodology of adsorbent Preparation



### 3.2.3 Preparation of stock solution of arsenate water

As(V) stock solution was made by dissolving potassium arsenate monobasic (H<sub>2</sub>AsKO<sub>4</sub>) (Sigma Aldrich, USA) in distilled water. For preparation 50ppb of As(V) initially prepare 1000ppm of As(V) so take 0.24029g of Potassium arsenate monobasic (H<sub>2</sub>AsKO<sub>4</sub>) put it in round bottle flask of 100 ml. Fill the flask at 100 ml mark with distilled water. Shake it by hand for 5 minutes. Add 1 drop of Nitric acid (HNO<sub>3</sub>) and shake it for 2 minutes. For 100ppm take 10ml of 1000ppm concentrated solution of As (V) and diluted into 100ml distilled water to make 100ppm As (V) concentrated solution. For 1ppm reduce the solution concentration of 100ppm to 1ppm. Take 1ml of 100ppm concentrated solution of As(V) and diluted into 100ml distilled water to make 1ppm or 1000ppb As (V) concentrated solution. For 50ppb reduce the solution concentration of 1ppm to 50ppb. Take 2.5ml of 1ppm concentrated solution of As(V) and diluted into 100ml distilled water to make 0.05ppm or 50ppb As(V) concentrated solution.



**Figure : 3.2** Preparation of Stock Solution

### **3.3 Preparation of adsorbent**

#### **3.3.1 Cutting and washing of PAN fiber**

The Polyacrylonitrile fiber was cut into small pieces. And then Polyacrylonitrile fiber washed with deionized water for remove the impurities attached on the surface of fiber. After washing Polyacrylonitrile fiber keep for drying at room temperature overnight.

#### **3.3.2 Chemically modification of PAN fiber**

The PAN fiber was chemically modified using Dimethylacetamide ( $C_4H_9NO$ ), and aluminum chloride hexahydrate ( $AlCl_3 \cdot 6H_2O$ ). For chemically modified of PAN fiber was taken 10 g of PAN Fiber and 250mL of Dimethylacetamide (>97%) and used 5 g of Aluminum chloride hexahydrate as a catalyst in a 500mL round bottom flask. The flask was equipped with heating mantle and a reflux condenser mounted on magnetic stirrer plate as shown in Fig. 3.4. PAN fiber was heated for 8 hours at  $90^\circ C$  and stirred at 340 rpm. The fiber was Modified using  $C_4H_9NO$  in the presence of ( $AlCl_3 \cdot 6H_2O$ ) catalyst. This reaction in heating mantle was continuously monitored and temperature and mixing were controlled. After 8 hrs. heating, PAN fiber was cooled to room temperature, then pH was checked and found 13. The PAN fiber samples were washed with deionized water (D.I) repeatedly and the pH after washing with D.I water was 10. PAN fiber samples were dried at room temperature. PAN fiber samples were dried for the iron loading by precipitation method.



**Figure: 3.3** Modification of PAN Fiber

### **3.3.3 Iron loading**

Modified PAN Fiber will be immersed in the 250 mL  $\text{FeCl}_3$  solution. Modified PAN Fiber will be stirred gently at room temperature for 3 hours. After stirring for 3 hours, 0.5 M NaOH solution (~200 mL) will then be added slowly dropwise to the Modified PAN–iron mixture while stirring. pH will be maintained at 8–10 using pH strips. The mixture will then be stirred continuously for 1.5 hours.

#### **3.3.3.1 AGING**

The mixture will be left undisturbed for 12 hours at room temperature overnight to improve coating stability and crystallinity.

#### **3.3.3.2 WASHING**

Iron-coated PAN Fiber will be washed 5–6 times with ~150 mL distilled water for each wash. This process will continue until the wash water reaches neutral pH (~7).

#### **3.3.3.3 DRYING**

Iron-loaded Modified PAN Fiber will be dried in the oven for 6 hours and then left overnight at room temperature. After this, it will be fully prepared to use as an adsorbent.

### 3.3.4 Filtration assembly

The whole filtration process is carried out in the column which is available in the water quality laboratory which has internal diameter of 0.5 cm and has height of 14cm the height of the Modified PAN Fiber depend on the height of the column is made up of borosilicate glass which has inlet for the feed introduced at the top and the treated water is collected from the bottom of the column it also has the valve named regulator valve which help out to maintaining the residence contact time this whole assembly is attached with the peristaltic pump for the control of the feed flow. And then set the time, flowrate inlet feed concentration and bed height as shown in the table to analyses the adsorption study.



**Figure: 3.4** Iron Loaded Chemically Modified PAN Fiber Column Study

### 3.4 Adsorption study of the Chemically Modified PAN fiber

The column tests packed with modified PAN fiber was conducted for synthetic arsenate water at different operating parameters i.e, flowrate, bed height, initial concentration, and pH. The experiment conducted at given conditions of flow rate (5mL/min, 10mL/min, and 15mL/min), arsenic contaminated water was flow over different bed height of modified PAN fiber (1cm, 2cm, and 3cm), and fixed concentration of synthetic water As(V) (30mg/L, 40mg/L and 50mg/L) and the pH of arsenic contaminated water (2, 7, and 10) as shown in table 3.2.

**Table 3:2** Column study Parameters

| Synthetic Arsenate water     |    |                   |                 |            |
|------------------------------|----|-------------------|-----------------|------------|
| Arsenate Inlet Concentration | pH | Flowrate (mL/min) | Bed Height (cm) | Time (min) |
| 50ppb                        | 2  | 5                 | 1               | 50         |
|                              | 7  | 10                | 2               | 100        |
|                              | 10 | 15                | 3               | 150        |

### 3.4.1 Breakthrough curve

The performance of a fixed-bed column is described through the concept of the breakthrough curve. The time for breakthrough appearance and the shape of the breakthrough curve are very important characteristics for determining the operation and the dynamic response of an adsorption column. The loading behavior of arsenic to be adsorbed from solution in a fixed bed is usually expressed in term of  $C_t/C_0$  as a function of time or volume of the effluent for a given bed height, giving a breakthrough curve. The value of  $q_{total}$  for a given feed concentration and flow rate is equal to the area under the plot of the adsorbed As concentration  $C_{ad}$  ( $C_{ad} = C_0 - C_t$ ) (mg/l) vs.  $t$  (min) and can be calculated from Eq. (2):

$$q_{total} = \frac{Q \cdot A}{1000} = \frac{Q}{1000} \int_{t=0}^{t=t_{total}} C_{ad} dt \quad (2)$$

Where,  $t_{total}$  is the total flow time (min),  $Q$  is the flow rate (mL/min) and  $A$  is the area above the breakthrough curve,  $C_{ad}$  (mg/L) is the adsorbed concentration. The total amount of adsorbates flow through the column is calculated by Eq. (3):

$$m_{total} = \frac{C_o \cdot Q \cdot t_{total}}{1000} \quad (3)$$

The equilibrium uptake  $q_{eq}$  (mg/g) or maximum capacity of the column in the column is calculated by Eq. (4):

$$q_{eq} = \frac{q_{total}}{m} \quad (4)$$

where  $m$  is the dry weight of adsorbent in the column (g).

### 3.4.2. Adsorption isothermal study

Adsorption isotherm is the relationship between the adsorbate in the liquid phase and the adsorbate adsorbed on the surface of the adsorbent at equilibrium at constant temperature. The isothermal studies of adsorption play an important role in adsorption column. Three isotherm models, i.e. Langmuir, Freundlich and Dubinin - Radushkevich (D-R) were used to find out the isotherm constant and regression coefficient of adsorption test results. Isothermal experiments were conducted by changing initial As concentration from 12.5 to 500 µg/L, keeping adsorbent dose 0.5 g fixed. In isotherm experiments, equilibrium time was maintained at 2 hours and the pH was adjusted at 7 for As(V).

### 3.5. Methods of arsenic testing

#### ARSENIC REMOVAL (%)

The percentage Arsenic removal or percentage filtration is calculated by the formula that is.

$$\% \text{ Removal} = \frac{C_i - C_f}{C_f} \times 100$$

Where,  $C_i$  is the inlet concentration of the arsenic contaminated water and the  $C_f$  is the Arsenic concentration in the treated water at outlet.

### 3.6. Characterization Of Adsorbent

Characterization of adsorbent were carried out by different equipment in order to analyze the structure of adsorbent during different stages and its behavior at different wave number. It was carryout by the following methods:

1. Fourier-Transform infrared spectroscopy (FTIR)
2. Scanning Electron Microscopy (SEM)

#### 3.6.1 FTIR Spectrometry

Fourier Transform Infrared (FTIR) spectroscopy is an analytical technique that records spectra based on the interference of infrared light frequencies. The instrument consists of a source, beam splitter, mirrors, laser, and detector. Infrared light is split into two beams, one reflected by a fixed mirror and

the other by a moving mirror. After recombination, the beam passes through the sample, where absorption occurs, and the transmitted portion is detected. The resulting interferogram contains information about all wavelengths simultaneously, which is then digitized and processed (Thermo Fisher, 2022).

In this study, Polyacrylonitrile (PAN) Fibers were analyzed to identify functional groups using an FTIR spectrometer (Nicolet iS10, Thermo Fisher Scientific, USA) equipped with Attenuated Total Reflectance (ATR). Samples were dried in an oven (Lab Tech LDO-030N, Daihan Labtech Co., Ltd., Korea) at 70 °C for 5 hours to remove moisture. FTIR spectra were recorded in the range of 4000–60 cm<sup>-1</sup> at ambient room temperature for Un-Modified PAN fibers, Modified fibers, and fibers after adsorption, at Hitech Laboratories (IARSCS), University of Sindh, Jamshoro.

### **3.6.2 Scanning Electron Microscope (SEM)**

The scanning electron microscope (SEM) is an advanced imaging tool that uses electrons rather than light to generate high-resolution images of surface morphology. By directing a focused electron beam (0.5–30 kV) onto a sample, secondary and backscattered electrons are produced, which reveal fine topographical and compositional details. SEM offers excellent depth of field, spatial resolution below 10 nm, and sensitivity to surface features under 5 nm when operated at low beam energies. In addition to imaging, detectors for backscattered electrons and x-rays enable elemental and structural analysis, including 3D chemical surface mapping with magnifications up to 50,000×

In this study, the surface morphology of un-Modified and Modified Polyacrylonitrile (PAN) fibers, as well as fibers after adsorption, was examined using a JEOL JSM-6380 LV SEM (Japan) at the Mining Engineering Department laboratory, MUET, Jamshoro (Zhang et al., 2007).

## **CHAPTER 4**

### **RESULT AND DISCUSSION**

#### **4.1 OVERVIEW**

In this chapter, un-functionalized and functionalized (Polyacrylonitrile fiber) was used as an adsorbent to optimize the removal efficiency of arsenic from synthetic arsenate water.

Characterizations of adsorbent were conducted of un functionalized Polyacrylonitrile, chemically modified Polyacrylonitrile, iron loaded Polyacrylonitrile, and functionalized Polyacrylonitrile and after adsorption.

The evolution of functional groups, surface morphology and elemental analysis were studied using a Fourier Transform Infrared Spectroscopy (FT-IR) and Scanning Electron Microscopy (SEM) was conducted.

Initially un-functionalized Polyacrylonitrile fiber used as an adsorbent in fixed bed column for removal of arsenic from synthetic water. After functionalized used adsorbent in fixed bed column for optimized the adsorbent efficiency and then compared the results of both adsorbents. The fixed bed column study experiments were performed to optimize the various parameter i.e. Height of bed, flow rate, pH, and isotherm studies were also conducted.

#### **4.2 Characterization Of Adsorbent**

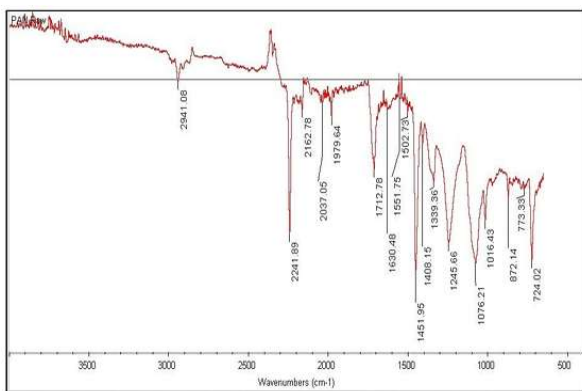
##### **4.2.1 Fourier Transform Infrared Spectroscopy (FTIR)**

In Hitech Laboratories (IARSCS), University of Sindh, Jamshoro, PAN fibers was analyzed to determine the presence of functional groups using FTIR spectrometry. Analysis of PAN samples with a range of 4000–60 cm<sup>-1</sup> was carried out using FTIR model number (Nicolet iS10, Thermo Fisher Scientific, USA) with Attenuated Total Reflectance (ATR).

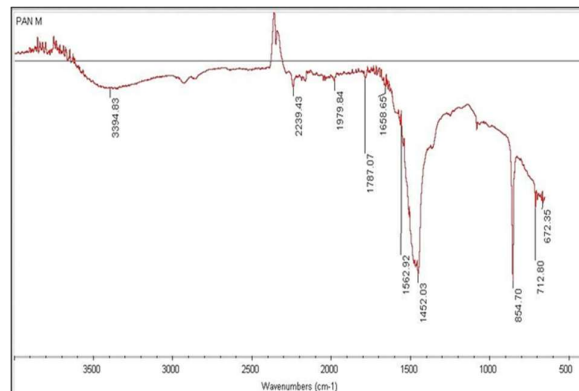
PAN fibers was chemically synthesized to attach functional groups, followed by the 55 precipitation of iron loading. Un-Modified Polyacrylonitrile (PAN), modified Polyacrylonitrile fibers (M-PAN), modified Polyacrylonitrile iron loading (M-PAN-Fe).

And arsenic adsorption were all studied using FTIR spectra. The correlation chart was used to provide identities to the FTIR spectra (Colthup, 1950).

The peak at 1245 cm<sup>-1</sup> (C-O), 1339 cm<sup>-1</sup> methyl group (CH<sub>3</sub>), 1451 cm<sup>-1</sup> methylene (CH<sub>2</sub> bending), 1712 cm<sup>-1</sup> carbonyl (C=O stretching), 2241 cm<sup>-1</sup> nitrile (CN), and 2941 cm<sup>-1</sup> aliphatic are visible in the un-Modified -PAN's FTIR spectrum in Fig. 4.1. (CH). This indicates that the PAN fibers is an acrylonitrile and methylacrylate copolymer. The aminated PAN fiber's spectrum, which has a peak at 2241 cm<sup>-1</sup>, is linked to the (CN) group. These findings are related to an FTIR analysis of nanofiber membranes composed of aminated polyacrylonitrile [27].

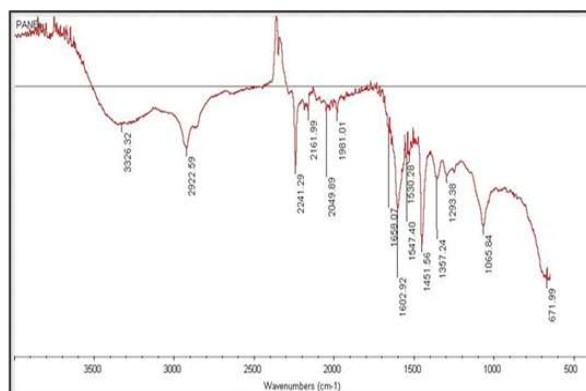


**Figure. 4.1:** FTIR of Un-Modified PAN Fiber



**Figure. 4.2:** FTIR of Chemically modified PAN

The FTIR spectrum of the modified PAN in Fig. 4.2 shows peaks at 1245 cm<sup>-1</sup> (C-O), 1339 cm<sup>-1</sup> methyl group (CH<sub>3</sub>), 1451 cm<sup>-1</sup> methylene (CH<sub>2</sub> bending), 1712 cm<sup>-1</sup> carbonyl (C=O stretching), 2241 cm<sup>-1</sup> nitrile (CN), and 2941 cm<sup>-1</sup> aliphatic (CH). This indicates that the PAN fibers is a copolymer of acrylonitrile and methylacrylate. The (CN) group is associated with the peak of the spectra of the aminated PAN fibers at 2241 cm<sup>-1</sup>. These results come from an FTIR examination of aminated polyacrylonitrile nanofiber membranes [28].



**Figure: 4.3** FTIR of Modified Iron Loaded PAN Fiber

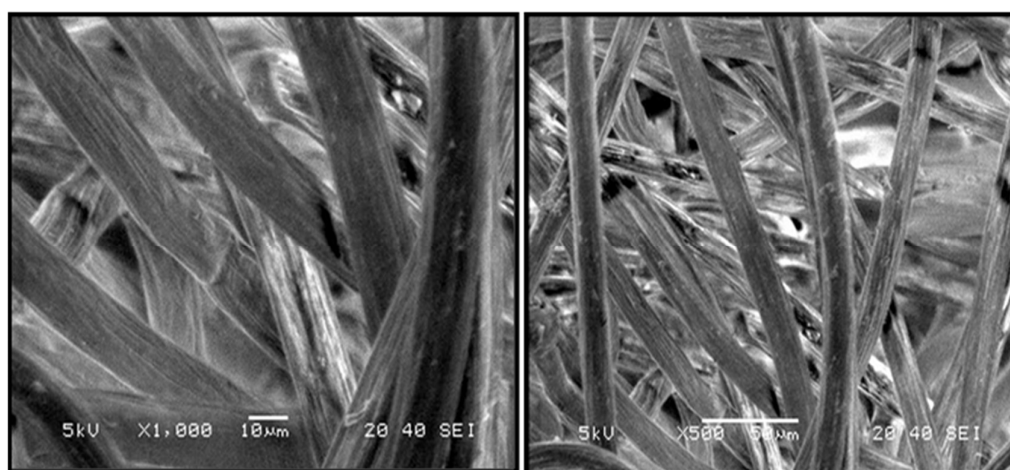
It is shown in Fig. 4.3 that the spectra of modified PAN with iron loading (MPAN-Fe) have a peak at 1451  $\text{cm}^{-1}$  ( $\text{CH}_2$  bending). The other peaks are at 1602  $\text{cm}^{-1}$  for  $\text{C}=\text{N}$ , 2241  $\text{cm}^{-1}$  for  $\text{CN}$ , 2922  $\text{cm}^{-1}$  for  $\text{CH}$ -aliphatic, 1658  $\text{cm}^{-1}$  for  $\text{NH}$ , and 3326  $\text{cm}^{-1}$  for  $\text{NH/OH}$ . Since the  $\text{NH}$  group in carboxamide is represented by the peak at 1658  $\text{cm}^{-1}$ , it is apparent that the iron was precipitated by neutralising  $\text{NaOH}$ . The fibers that has been exposed to  $\text{NaOH}$  displays a narrow band between 3,100 and 3,600  $\text{cm}^{-1}$  that corresponds to the hydroxyl group ( $\text{OH}$ ) in carboxylate and verifies the formation of carboxylate cation-exchange groups [29].

#### 4.2.2 Scanning Electron Microscopy (SEM)

At the Mining Engineering Department Laboratory, MUET, Jamshoro, the surface morphology of the samples, including un-Modified and Modified PAN fibers, and after adsorption, were examined using a scanning electron microscope (SEM) model.

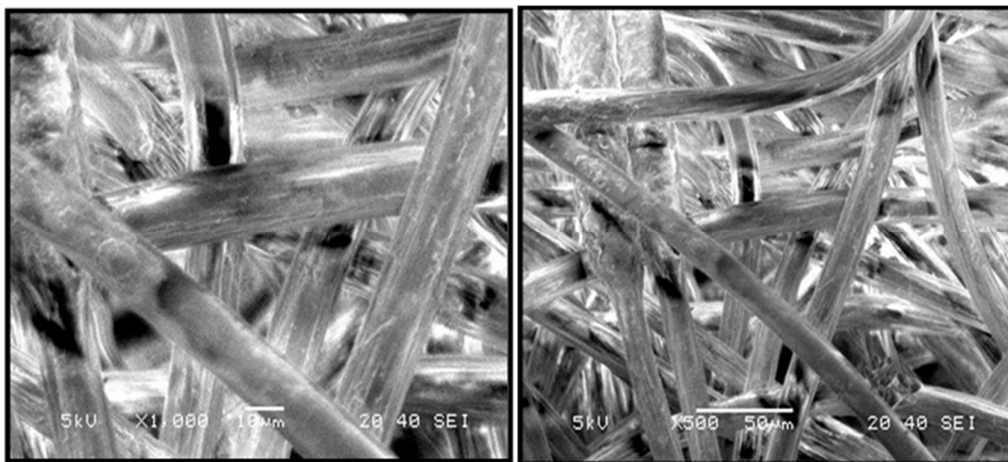
Analyses were conducted on the surface morphology of un-Modified Polyacrylonitrile (PAN), modified Polyacrylonitrile fibers (M-PAN), modified Polyacrylonitrile iron loading (M-PAN-Fe) and after arsenic adsorption.

The smooth and homogeneous surface morphology of un-Modified Polyacrylonitrile (PAN) is shown in Fig. 4.4.



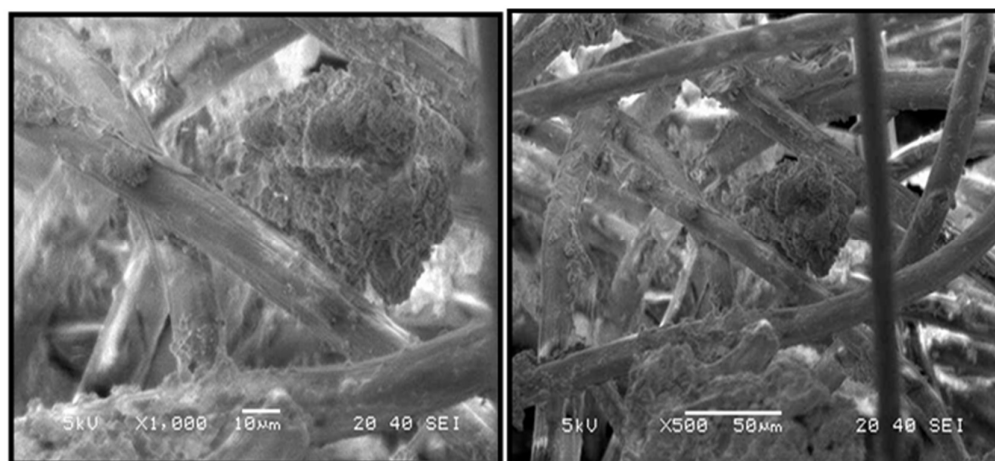
**Figure. 4.4:** SEM of Un-functionalized PAN Fiber

The chemically modification of PAN fiber with diethylenetriamine (MPAN) defines that the something attached on the surface shown in Fig. 4.5, and after modification surface pores became larger.



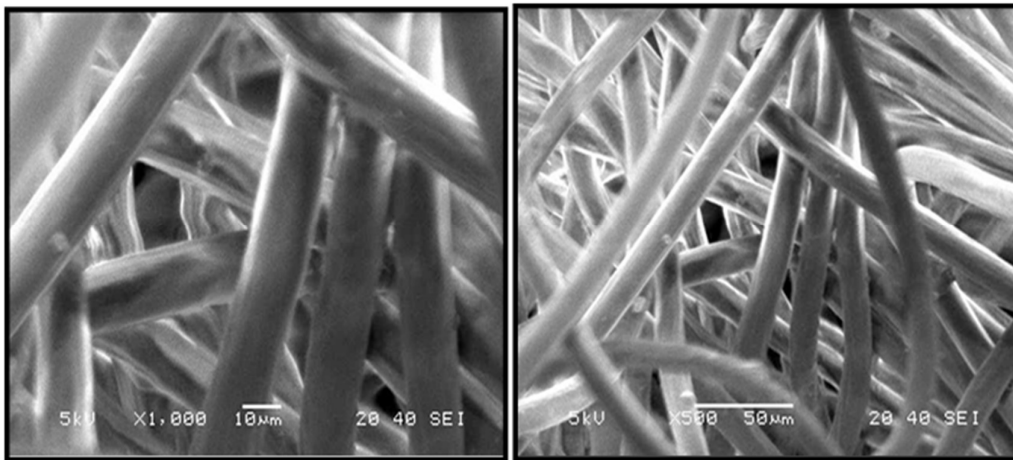
**Figure. 4.5:** SEM of Modified PAN Fiber

After chemically modification of M-PAN fiber loaded with iron (MPAN-Fe) defines that the surface spongy like rough and porous surface with some irregular shapes shown in Fig. 4.6.



**Figure. 4.6:** SEM of MPAN-Fe Fiber

Adsorbent became rough after the adsorption process seen in Fig. 4.7 was complete because the arsenic contained in the water sample was extracted and stuck to the surface of the adsorbent.



**Fig. 4.7** SEM of Modified MPAN-Fe Fiber After Adsorption

### **4.3 ADSORPTION STUDY**

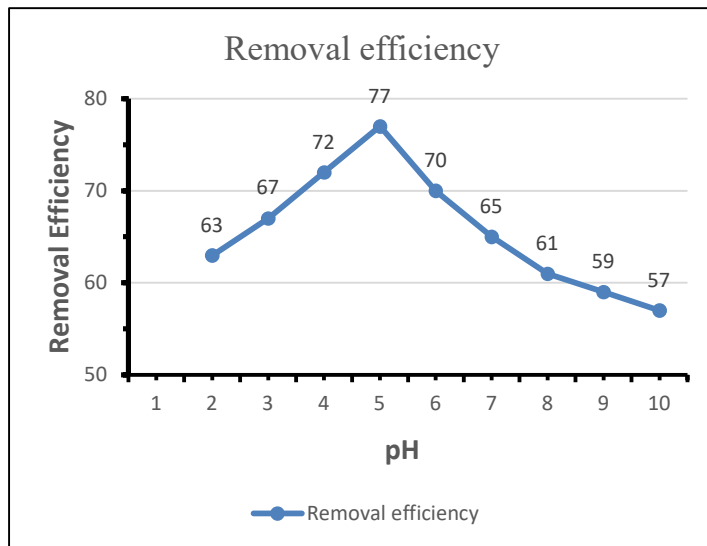
#### **4.3.1 Un-Functionalized Pan Fiber**

##### **4.3.1.1 Effect Of Ph On Synthetic Arsenate Water (As(V))**

As(V) on un-Modified PAN fiber adsorbent at different pH solutions (2 to 10) is a very essential characteristic, and this is depicted in Fig. 4.8. This was investigated to clarify how solution pH affects adsorption capacity and to establish the optimal pH for arsenic adsorption in various pH ranges.

The adsorption of As(V) onto PAN fiber adsorbent was investigated in the current work using 0.1 M HCl and 0.1 M NaOH solutions at various pH values ranging from 2 to 10. At room temperature (25 °C), 3 cm bed height, and flow rate of 5 mL/min with input initial concentration (50 g/L) of arsenic solution, experiments were carried out to investigate the influence of initial solution pH on the adsorption of As(V). As illustrated in Fig. 4.16, it was discovered that the adsorption effectiveness was 63% at pH 2, gradually climbed to 77% at pH 5, and then fell as pH increased, with a minimum adsorption efficiency of 57% at pH 10.

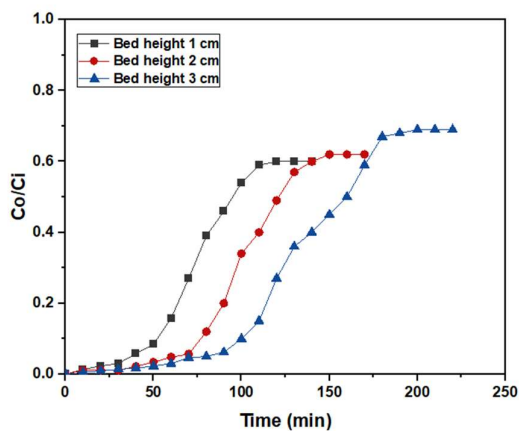
Depending on the reaction conditions, a significant electrostatic interaction between protonated amino and hydroxyl groups and negatively charged arsenate ions caused As(V) to adsorb onto un-functionalized PAN fiber. The amount of As(V) ions that can bind to surfaces will decrease as the pH rises because OH ions in the solution can outcompete arsenate ions. As a result, As(V) had its pH altered to be between pH 5 and 7 for all adsorption investigations.



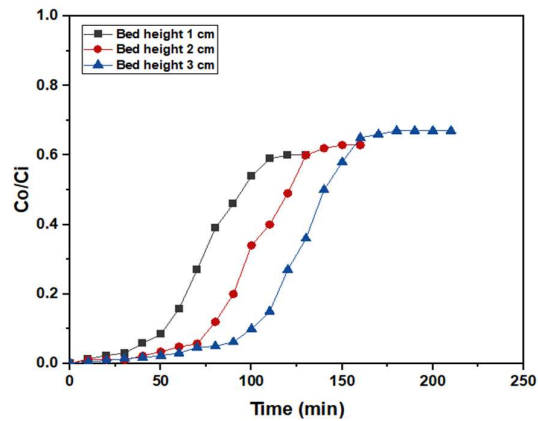
**Figure. 4.8:** Effect of pH on Synthetic Arsenate Water (As(V))

#### 4.3.1.2 Effect of Bed Heights Synthetic Arsenate Water

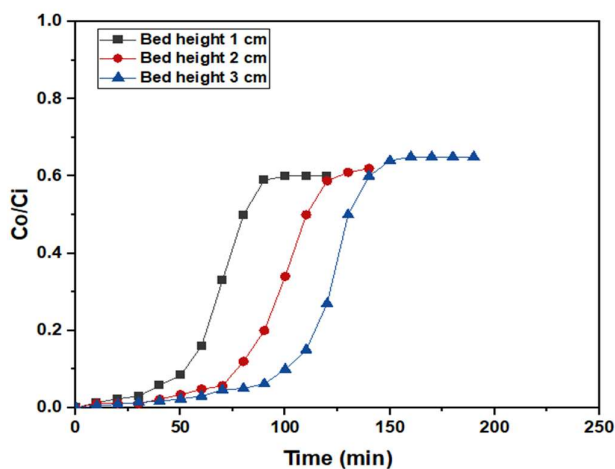
The breakthrough curve for As(V) adsorption on un-Modified PAN fiber was obtained for three different bed heights of 1, 2, and 3 cm at constant adsorbate feed flowrates of 5 mL/min, 10 mL/min, and 15 mL/min, and an adsorbate inlet concentration of 50 g/L and 5 pH (Fig. 4.9(a)), (Fig. 4.9(b)), and (Fig. 4.9(c)). Both the break-through time ( $t_b$ ) and exhaustion time ( $t_e$ ), as shown by the plots in Figure 4, were discovered to increase with rising bed height. The graphs show that with variations in bed depth, which is also predicted, the shape and gradient of the breakthrough curves were slightly different. Due to the increased amount of PAN fiber that allowed for more cation fixations and active binding sites for the adsorption process to take place, a greater uptake was seen at higher bed heights. The mass transfer zone has grown as bed height has increased. The mass transfer zone advanced in a column away from the bed's entrance and toward the door. Therefore, with the same influent concentration and fixed bed system, an increase in bed height resulted in a large distance for the mass transfer zone to travel, which in turn caused a prolonged breakthrough time. Since there was more adsorbent material present at greater bed heights, the treated solution's volume increased since the bigger service area was available.



**Figure : 4.9 (a)** Feed flowrate 5 ml/min



**Figure : 4.9(b)** Feed flowrate 10



**Figure : 4.9 (c)** Feed flowrate 15

**Figure. 4.9:** Effect of adsorbent bed height (1cm, 2cm, and 3cm) and sampling time from 0 to 250 mints arsenic removal efficiency were examined at flow of (a) 5mL/min, (b) 10mL/min and (c) 15mL/min feed flow rate.

#### 4.3.1.3 Effect of Flowrates Synthetic Arsenate Water

As shown by the breakthrough curve in (Fig. 4.10(a)), (Fig. 4.10(b)), and (Fig. 4.10(c)). the effect of feed flowrates on the adsorption of As(V) on un-modified PAN fiber was investigated by varying the feed flowrates of 5, 10, and 15 mL/min with constant adsorbent bed heights of 1 cm, 2 cm, and 3 cm, as well as an inlet adsorbate concentration of 50 g/L and at pH 5. The graph demonstrated that the column became saturated earlier at higher flowrates when the front of the adsorption zone soon approached the top of the column. Because of the lower flowrate, the adsorption zone is shallower and the contact period is longer. Less adsorption absorption was seen at higher flowrates due to a steeper curve with a very early breakthrough and exhaustion period. Due to the longer contact time between the adsorbent and influent, the highest efficiency is attained at a flowrate of 5 mL/min.

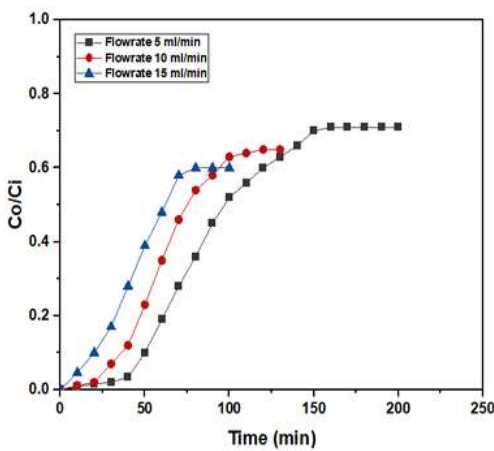


Figure: 4.10 (a)

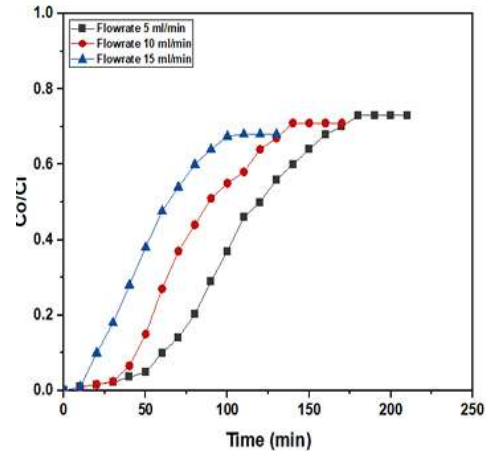


Figure : 4.10 b

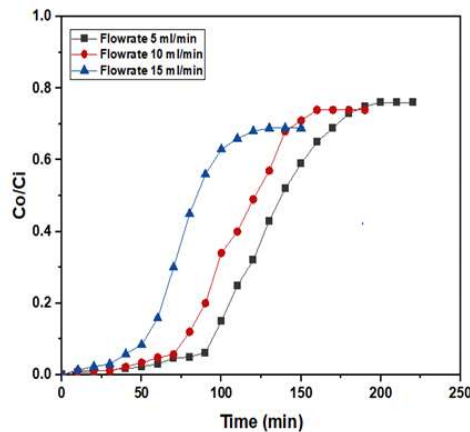


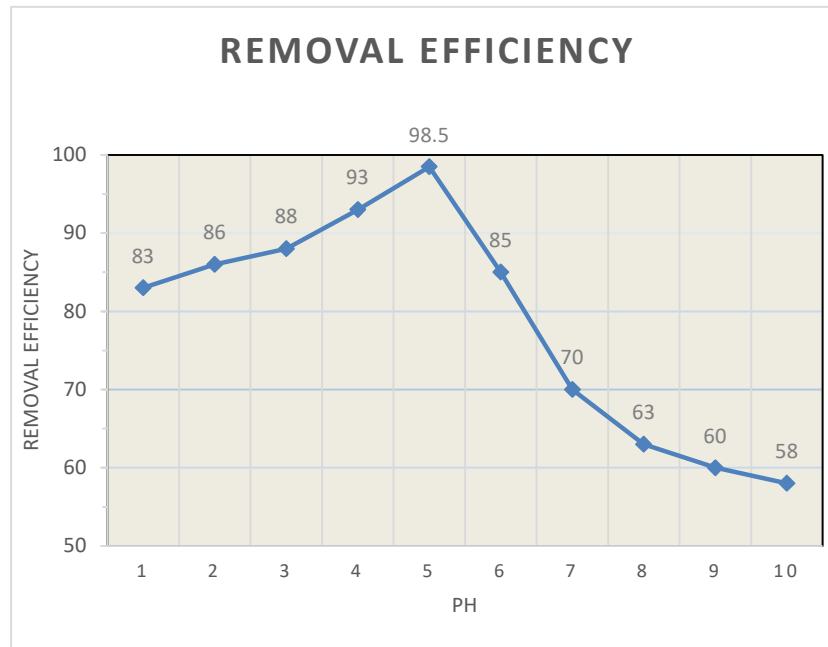
Figure: 4.10(c)

**Figure:4.10** Show The Effect Of Absorbent At Different Flow Rates (5ml/Min, 10ml/Min And 15ml/Min ) And Sampling Time From 0 To 250 Mints Arsenic Removal Efficiency Were Examined At Bed Height 1cm, 2cm And 3cm.

## 4.3.2 Functionalized Pan Fiber

### 4.3.2.1 Effect of pH on Synthetic Arsenate Water (As(V))

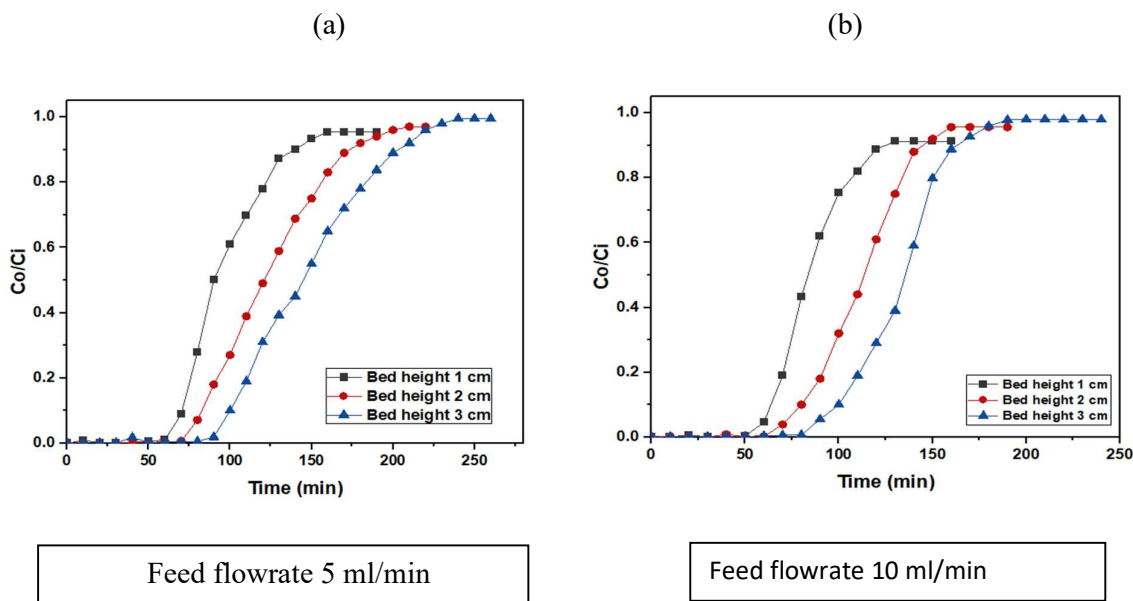
Using 0.1 M HCl and 0.1 M NaOH solutions, the impact of the initial solution pH on the adsorption of either As(V) in the investigated column was examined at pH 2 to 10 (representing acidic, neutral, and alkaline conditions, respectively), while maintaining the adsorbent bed height at 3 cm, initial adsorbate concentration at 50 g/L, and influent flow rate at 5 mL/min at room temperature (25.1°C). The initial treatment The ionic states of the functional groups of the adsorbent and metal species in the solution are affected by pH, making it one of the most significant experimental parameters for the adsorption process [30]. When the pH was between 2 and 4, the removal efficiency improved dramatically, and at pH 5, the highest removal efficiency was reached when the pH was between 6 and 10. the removal efficiency seen in Fig. 4.11 decreased significantly. High pH reduces the adsorption-driving forces between  $\text{H}_3\text{AsO}_4$  and the adsorbent and leads to reduced adsorption. The  $\text{H}_2\text{AsO}_4$  species predominates in the pH range of 2 to 4. The greatest electrostatic contact between  $\text{H}_2\text{AsO}_4$  and protonated-amino groups on the surface of functionalized PAN fibre occurs at pH 5, when the concentration of  $\text{H}_2\text{AsO}_4$  is at its highest and the removal efficiency approaches 98.5%. As the pH rises,  $\text{OH}^-$  ions in the solution can compete arsenate ions, resulting in less As(V) ion adsorption.

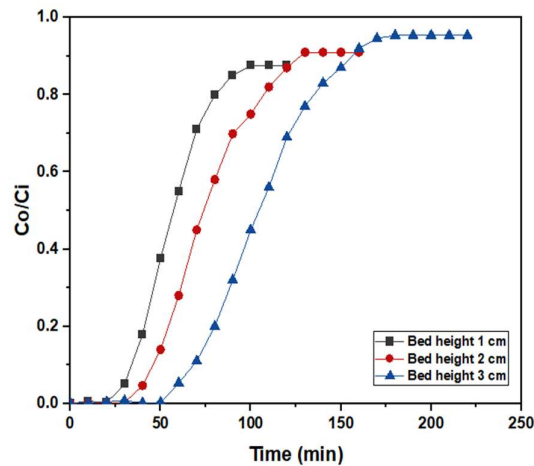


**Figure. 4.11:** Effect of pH on Synthetic Arsenate Water (As(V))

#### 4.3.2.2 Effect of Bed Heights on Synthetic Arsenate Water

The (Fig. 4.12(a)), (Fig. 4.12 (b)) and (Fig. 4.12(c)) presents the breakthrough curves for different adsorbent bed heights of 1, 2, and 3 cm at an As(V) concentration of 50 g/L, an initial solution pH of 5, and flow rates of 5 mL/min, 10 mL/min, and 15 mL/min, respectively. The saturation point was steadily attained after crossing the breakthrough point, as evidenced by the comparable appearance of the breakthrough profiles of As(V) adsorption and their observation to have reduced inclination at greater bed heights. The breakthrough time for As(V) adsorption increased with the bed height increasing from 1 to 3 cm, and the saturation time for As also increased (V). Greater mass of the adsorbent in a bed with a higher height takes longer to reach the breakthrough and saturation stages. Longer saturation times, according to Jang and Lee (2016), may help ensure that As ions contact with the adsorbent enough, increasing the adsorption capacity. The saturation adsorption capacity of the adsorbent As(V) adsorption, however, decreased in this investigation as the bed height increased from 1 to 3 cm. The decrease in saturation capacity at a high bed height is most likely caused by As ions' inaccessibility to all of the adsorbent's active sites [31,32] This revealed that improving adsorption efficiency by utilising a relatively low bed height of the adsorbent in the column is preferable to employing a greater bed height of adsorbent in some cases.



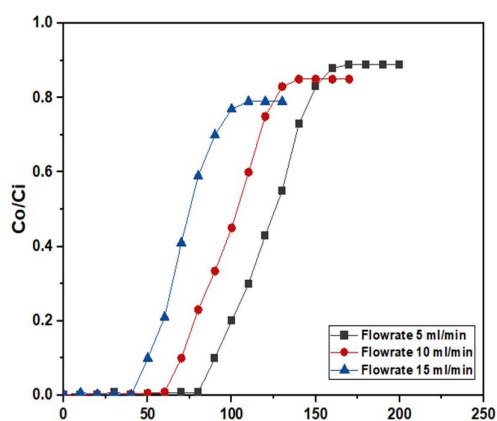


(c)

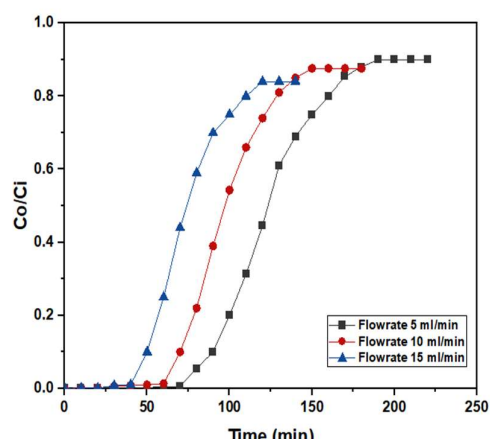
**Figure. 4.12:** Effect of adsorbent bed height 1cm, 2cm, and 3cm and sampling time from 0 to 250 mints arsenic removal efficiency were examined at flow of (a) 5mL/min, (b) 10mL/min and (c) 15mL/min feed flowrate.

#### 4.3.2.3 Effect of Flowrates on Synthetic Arsenate Water

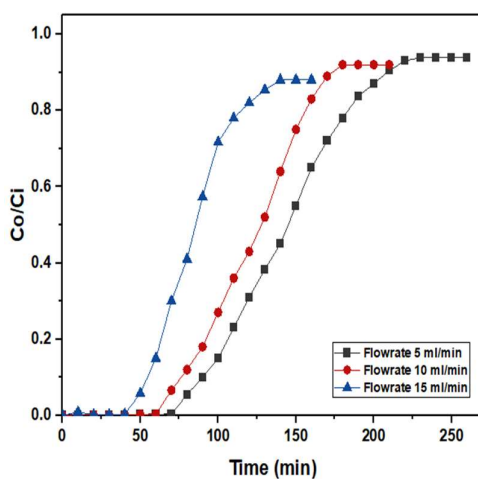
When As(V) solution had an initial concentration of 500 g/L, pH was 5, and the adsorbent bed height was 1 cm, 2 cm, and 3 cm, respectively, the influence of flow rates on As(V) adsorption in the continuous fixed-bed column packed with Modified PAN fiber was investigated. The breakthrough curves at lower flow rates resembled more gradual decline curves due to When(V) adsorption, and they became noticeably steeper as the flow rates were raised (Fig. 4.13 (a)), (Fig. 4.13 (b)) and (Fig. 4.13 (c)). The breakthrough time of When(V) and the saturation time of As(V) both decreased as the flow rate rose from 5 to 15 mL/min. Slower flow rates resulted in a progressive occupancy toward the breakthrough and saturation points because the adsorbate ions remained in the column system for a longer period of time, allowing them to more effectively interact with the adsorbent's binding sites. Higher flow rates result in inadequate residence durations for either better metal ion interaction with active sites or diffusion into internal pores [33]. which causes the breakthrough profiles to have a steeper slope. The saturation adsorption capacity of As(V) decreased with an increase in flow rate. Because injecting the influent into the column at a rapid velocity causes the solute to depart before reaching equilibrium and lowers the adherence of the adsorbates to the adsorbent, higher flow rates resulted in lower adsorption capacity [34].



(a) Bed height 1



(b) Bed height 2



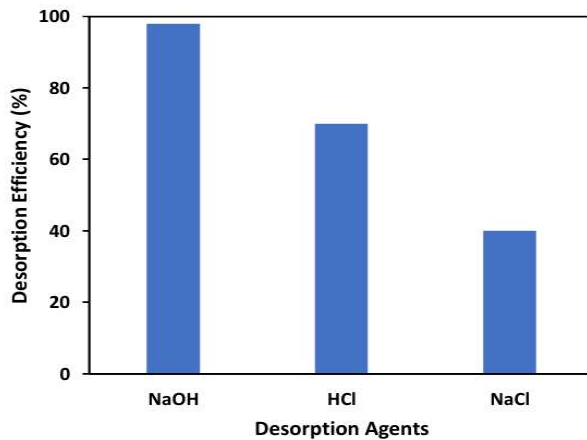
(c) Bed height 3 cm

**Figure. 4.13.** Effect of adsorbent Flow rates 5mL/min, 10mL/min and 15mL/min and sampling time from 0 to 250 mints arsenic removal efficiency were examined at bed height (a) 1 cm, (b) 2cm and (c) 3cm

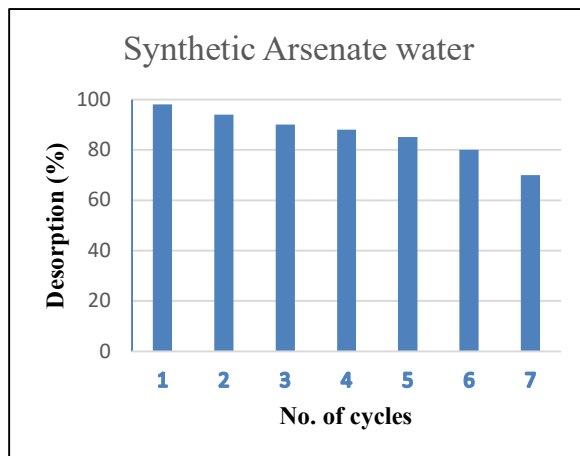
#### 4.4 DESORPTION STUDY

Figure 4.16(a) shows the ability of Modified PAN Fiber to regenerate following As(V) adsorption using different desorbing agents (NaOH, NaCl, and HCl) at concentrations of 0.5 M. The best outcomes were obtained using 0.5 M of NaOH for a contact time of 12 h, allowing the recovery in water of >90% of adsorbed As(V). The adsorbent was removed from the solution, cleaned with distillate water, dried at 50 C, and then used for As(V) adsorption following the desorption experiment with 0.5 M of NaOH. The adsorption-desorption cycles of Modified PAN Fiber were depicted in Fig. 4.16(b). It is clear that after five cycles, the synthetic arsenate water's As(V) removal efficiency was maintained at around 90%.

Following adsorption-desorption cycles, the clearance % sharply dropped. The MPAN-Fe surfaces' inability to completely desorb As(V) and synthetic arsenic may be to blame for the buildup. The MPAN-Fe- is effective for three adsorption-desorption cycles, according to the data, and is reusable.



**Figure: 4.14(a):** Desorption Efficiency of Arsenic using different desorption Agents



**Figure: 4.14(b):** Adsorption–desorption of As ions and reusability of adsorbent using NaOH as desorbing agent

## 4.5 EFFICIENCY COMPARISON

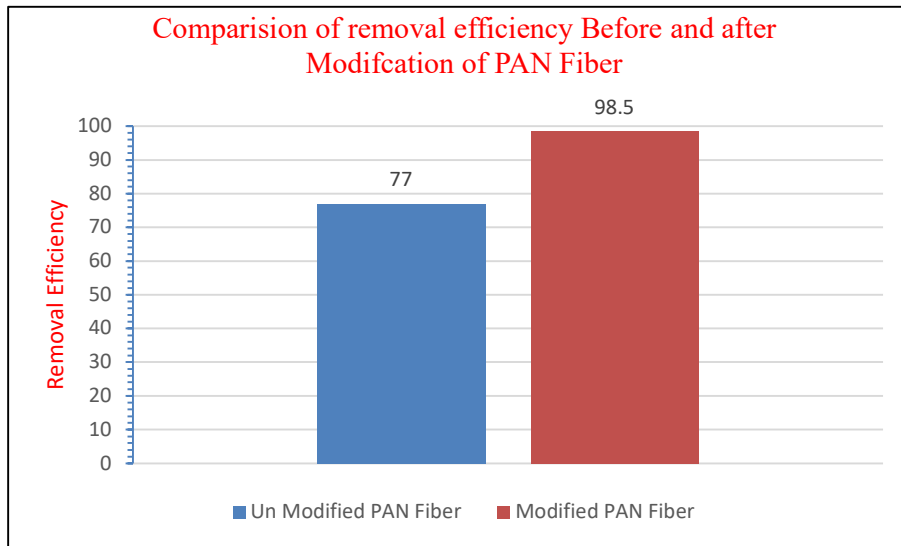
The removal efficiency comparison un-Modified PAN fiber and Modified PAN fiber is given below:

### 4.5.1 Removal Efficiency Comparison with Un-Modified and Modified PAN Fiber of synthetic arsenate water.

The comparison of adsorption removal efficiency between un-Modified and Modified PAN fiber are shown in Table 4.1. the removal efficiency of Un Modified PAN Fiber is low and high removal efficiency achieved of Modified PAN Fiber of adsorption of Arsenate (As(V)) shown in Fig. 4.16.

**Table 4.1:** Comparison Study of Adsorption Removal Efficiency of As(V)

| Synthetic arsenate water As(V) |      |
|--------------------------------|------|
| Un Modified PAN fiber          | 77   |
| Modified PAN fiber             | 98.5 |



**Fig. 4.15:** Comparison of Removal Efficiency As(V)

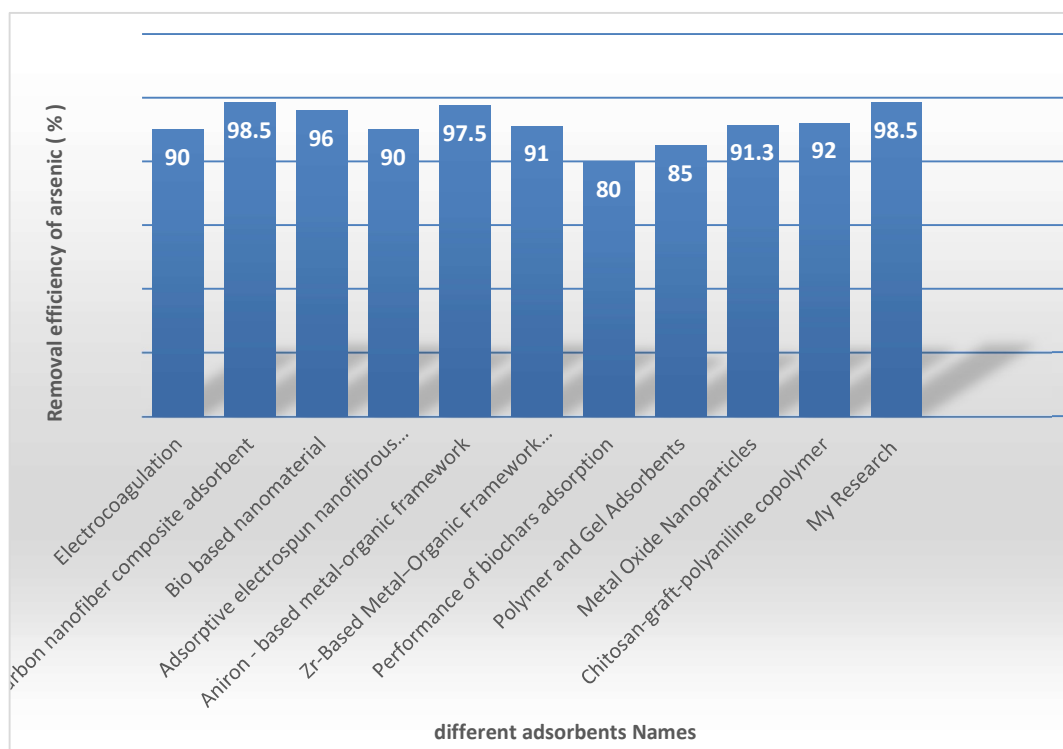
#### 4.6 Efficiency Comparison with previous research

Comparative Analysis of Other Adsorbents in Table 4.2 the greatest adsorption removal efficiency discovered during the current investigation is compared to the various adsorbents published in the literature for their capability to eliminate arsenic from groundwater. The greatest adsorption efficiency obtained in the current study is much higher than that discovered in previously published investigations, as can be seen in the table. It is generally easy to create large amounts of the Modified PAN fiber that was used in this experiment, and it is also relatively inexpensive. As a result, we can say that this iron loaded adsorbent may be employed as a cheap adsorbent to remove arsenic which is significant as illustrated in Fig. 4.17.

**Table 4.2 :** Different Adsorption technologies and removal Efficiencies.

| s.no | Different adsorption technologies                                            | Removal efficiency | Reference |
|------|------------------------------------------------------------------------------|--------------------|-----------|
| 1    | arsenic removal from synthetic water by electrocoagulation                   | 90                 | [1]       |
| 2    | carbon nanofiber composite adsorbent                                         | 98.5               | [2]       |
| 3    | Bio based nanomaterial                                                       | 96                 | [10]      |
| 4    | adsorptive electro spun nanofibrous membrane of polyacrylonitrile/organoclay | 90                 | [26]      |
| 5    | Aniron - based metal-organic framework                                       | 97.5               | [35]      |
| 6    | Zr-Based Metal–Organic Framework Nanoparticles                               | 91                 | [36]      |
| 7    | Performance of biochars adsorption                                           | 80                 | [37]      |
| 8    | Polymer and Gel Adsorbents                                                   | 85                 | [38]      |
| 9    | Metal Oxide Nanoparticles                                                    | 90.3               | [38]      |
| 10   | chitosan-graft-polyaniline copolymer                                         | 92                 | [39]      |

#### 4.6 Comparison of removal efficiency of Arsenic by different previously reported adsorbents



**Figure. 4.16:** Comparison of Removal Efficiency of Arsenic with previously different adsorbent

## **CHAPTER 5**

### **CONCLUSION AND RECOMMENDATION**

#### **5.1 CONCLUSION**

Arsenic removal from synthetic water using a fixed bed column is demonstrated in the current work. Under the right experimental circumstances, Modified PAN fiber works well as an adsorbent. Following are specific inferences made from the study's findings:

- The un- Modified PAN (raw) fiber was modified with dimethylacetamide ( $C_4H_9NO$ ) for 8 hours at 90 °C and stirred at 340 rpm.
- Modified PAN Fiber was immersed in the 250 ml  $FeCl_3$  solution. Modified PAN Fiber was Stirred gently at room temperature for 3 hours. After stirring 3 hours the then Slowly dropwise 0.5 M NaOH solution (~200 mL) was added to the Modified PAN–iron mixture while stirring.
- pH was maintained up 8-10 using pH strips.
- And then 1.5 hours stirred continuously.
- The surface modification of PANF, consisting in the conversion of the fibers nitrile groups into amino/acetamido group, produced a new efficient adsorbent for As(V) removal from synthetic water and arsenic from groundwater.
- Characterization of un- Modified PAN, MPAN and MPAN-Fe was performed by FTIR and SEM techniques. FTIR results revealed that MPAN-Fe fiber has more functional groups for adsorption of A(V).
- The surface morphology and elemental compositions of the PAN fiber were revealed by the SEM findings. The SEM results demonstrated that compared to Un- Modified PAN, MPAN, and MPAN-Fe, had the highest level of arsenic adsorption on its surface.
- The adsorption of As(V) onto adsorbent was studied at different pH values varying from 2 to 10, using 0.1 M HCl and 0.1 M NaOH solutions. The maximum removal efficiency was achieved at pH 5 due to strong electrostatic attraction between protonated amino and hydroxyl groups and negatively charged arsenate ions depending on the reaction conditions. At higher pH, OH ions in the solution can compete with the arsenate ions and so adsorption of As(V) ions will be reduced.

- Continuous adsorption method was used to study the As(V) removal by MPAN-Fe. Arsenic adsorption in the fixed bed column with this adsorbent had breakthrough curve profiles while varying different adsorption parameters, such as Initial concentration of arsenate As(V) (50 µg/L). Flowrate (5, 10 and 15 mL/min), Bed Height (1 (0.40g), 2 (0.75g) and 3 cm (1g)), and pH (2 to 10).
- Un-Modified PAN Fiber achieved maximum As(V) removal efficiency from synthetic water up to 76.2% with the optimum adsorbent bed height of 3 cm, pH 5, and 5mL/min at room temperature because of more contact time with adsorbent.
- Modified PAN Fiber achieved maximum (As) removal efficiency from synthetic water up to 99.5%. Which means Modified PAN Fiber is a sufficient adsorbent for (As) removal.
- The three different (NaOH, NaCl and HCl) desorption agents were used for desorption/regeneration of adsorbent. Among them, NaOH was chosen as the 94 best desorbing agents for Arsenic, due to their high desorption efficiency. The desorption efficiency of NaOH for As(V) Synthetic was about 98% and 88%, respectively. The exhausted adsorbent was regenerated up to nine cycles, and the removal efficiency of As(V) and As from groundwater on the recycled MPAN-Fe could be maintained at 76.1% and 68%, respectively, in the ninth cycle.
- The functionalized adsorbent was proven to be a reliable, affordable, and effective adsorbent for the removal of arsenic from household water in rural settings.

## 5.2 FUTURE WORK

The following directions could be taken with the research:

- Functionalized PAN fiber may also be used to remove additional harmful elements including lead, mercury, fluoride, and others to assess how well this adsorbent removes these metals.
- By expanding the size of the column and bed, the adsorption bed may be scaled up.
- The arsenic-affected districts of Sindh, such as Hala, Sakrand, Khairpur, Nawabshah and others, should have a pilot plant for arsenic treatment filters developed. This plant should incorporate iron ore or PAN fiber as an adsorbent.
- Government and non-governmental organisations should step forward and offer financial assistance to residents in arsenic-affected regions so they may purchase arsenic treatment facilities at a reasonable price.

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